

Deformation Operator Playbook

A methodology for controlled figure warps with safety locks and continuity

Anchors → Select → Transforms → Constraints → Viewfinder



Practical framework for operators for repeatable figure edits, without style fluff.

Executive Summary

Deformation Operator Playbook is a practical way to make image models do **repeatable figure edits**, on purpose, not by luck. Instead of vague style prompts, it uses a small set of operators (coil, parabolic stretch, depth tug, logarithmic scaling, etc.) written in a simple flow: **Anchors → Select → Transforms → Constraints → Viewfinder**. Anchors fix reference points, Select defines the region, Transforms apply the geometric move, Constraints keep it believable (thickness/topology/continuity locks), and Viewfinder frames the result without adding new distortion if desired.

Why it matters: most models “pretty up” anatomy rather than obey structure or authorship. This playbook gives you clear knobs and rails so a request like “the arm itself coils” stays anatomical with no ropes, no props, no sausage bulges. It’s fast to learn, audit-friendly, and portable across engines that respect literal instructions. Use it when you need edits that read as the body, not as an overlay: clean, measurable, teachable.

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Introduction

This is an experimental methodology for achieving controlled anatomical distortion in AI-generated images, a problem that currently has few established solutions. While the mathematical terminology (inverse-square fields, Fibonacci spirals, C2 continuity) provides systematic structure, current image models don't execute coordinate mathematics. Instead, this structured vocabulary functions as a translation protocol between mathematical reasoning systems (GPT) and pattern-matching visual systems. The math is a **translation layer**, not literal math control.

The AI image generation industry has oriented entirely around correcting anatomical distortions rather than controlling them intentionally. This closes off vast expressive territories that art has explored for centuries, from Giacometti's impossible elongation to Bacon's flesh distortions. This attempts to systematically reopen this creative space.



Example: Native engine → 1x logic applied → 2x logic applied

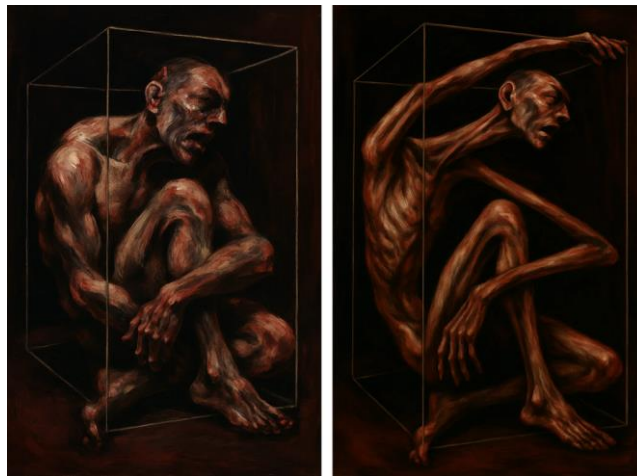


Fig: Native Engine → Multiple deformations 1x logic applied

The **Deformation Operator Playbook** is a practical toolkit for applying controllable distortions inside generative image systems. It gives a language for bending, extending, coiling, or modulating forms while keeping them coherent and integrated within the image.

With this playbook:

- **Extend or rotate limbs** along defined vectors and anchors.
- **Spiral or arc forms** to create graceful contradictions in anatomy.
- **Apply rhythmic modulations** (sine waves, parabolas) for serpentine or elastic effects.

- **Shift the viewfinder** to explore ghost density, framing tension, and compositional pull.

Each capsule is written in a compact, repeatable format: *Style* → *Anchors* → *Select* → *Transforms* → *Viewfinder*. This makes it easy to design single moves or stack multiple operators into compound transformations. The system is built for **artists, researchers, and prompt designers** who want more than chance: a way to script transformations, test them, and build images with structural consequence rather than surface novelty.

- **Who it's for.** Artists, researchers, and prompt designers who want more than chance, a way to script transformations, test them, and build images with structural consequence rather than surface novelty.
- **What it's not.** Not a direct geometric control system for today's text-to-image models; the math is a prompting scaffold, not literal execution. Evaluate it as a systematic prompting protocol, not a CAD tool.
- **Strengths.** Repeatability (parameterized operators), cross-engine transfer (swap adapters, keep logic), and teachability (results are inspectable; failures are legible).
- **Known friction.** Model priors may "normalize" small amplitudes; ambiguity can be literalized (coil → rope) without negatives and invariants; over-stabilization can hide failure evidence, document it.

Rather than prompting for stylistic effects or exploiting accidental glitches, this framework treats distortion as a separate mechanical layer that can be integrated seamlessly or continually grown.



Example: Native engine → 1x logic applied → 2x logic applied

Validation Approach

This work should be evaluated as experimental methodology for uncharted territory, not as theoretical precision about AI internals. Success is measured by whether it reliably produces intended distortions across multiple AI systems, regardless of whether the underlying mathematical constructs reflect actual model behavior.

Honest Limitations

- Mathematical terms are scaffolding for systematic exploration, not literal geometric execution
- This is systematic prompting, not a geometry engine.
- Reproducibility is relative to stochastic systems, not absolute
- Results depend on each model's pattern associations and training biases
- Engine drift / "collapse" and platform unpredictability or updates require ongoing retuning
- This is advanced prompting protocol, not direct geometric control

Bottom line. This is steering as structured geometry + composition + constraints, not "vibes + CFG." It bridges reasoning and rendering with a clear operator grammar you can measure, teach, and reuse.

Deformation Operator Playbook:

A Research Methodology for Purposeful Anatomical Distortion

The Unexplored Problem

Creating controlled, graceful anatomical distortions in AI-generated images is genuinely difficult. Most distortion work relies on accidental latent space artifacts or post-processing or are just coherently applied across multiple factors (such as all limbs) of the image so AI can “understandably” generate that fits within the defaults. Getting AI systems to generate purposeful and isolated anatomical impossibilities - like a spiraling arm that maintains anatomical coherence - during initial generation is uncharted territory and can be difficult to achieve.



Figs: Glitched → Predictable Distortion → Failed Distortion

What This System Actually Is

The Deformation Operator Playbook is a **research methodology** for systematically exploring controlled anatomical distortion. It functions as a conceptual scaffold that helps navigate an area where few established methods exist.

The core insight: treat distortion as a separate mechanical object, then instruct the system to integrate it seamlessly. This sidesteps AI systems' tendency to "correct" anatomical impossibilities by making the impossibility explicit and controlled.

The Framework's Function

This playbook works by creating "constructs of logic" - structured approaches to complex spatial transformations that are otherwise difficult to prompt consistently:

- **Separation of concerns:** Mechanics of deformation vs. aesthetics of integration
- **Permission structure:** Gives AI explicit license to break anatomy in specific ways
- **Systematic vocabulary:** Provides consistent language for complex spatial relationships
- **Cross-engine transferability:** Same logical structure works across different AI systems, especially instruction-following bias engines such as GPT and Gemini, while it still produces unpredictable results in aesthetic prior > literal geometry engines such as MidJourney and OpenArt. → *See appendix 2 for MJ and OA instructions*



Key Operational Patterns

The system extracts reliable prompt patterns like:

- "Extend the arm along this vector by a factor of $1.8\times$ while maintaining the elbow anchor"
- "Apply logarithmic scaling to this region while keeping proportional relationships intact elsewhere"
- "Create a 30-degree rotational transformation around this anchor point"
- "The figure's legs are a distinct, sculpted coil that spirals away from the body with the feet at the end"
- "Sculpted coil that spirals with the texture and legs twisting along, which must be in proportion to the body"

These extracts can then be applied to compositional work for increased mood and dramatic effect.



These work not because AI systems implement coordinate mathematics, but because the structured language creates consistent conceptual frameworks for the models to follow.

Integration with Mathematical AI and Why this isn't Literal Math

The mathematical language guidance serves multiple purposes:

- Provides structure that GPT-class systems can manipulate and refine
- Enables systematic generation of prompt variations
- Creates reproducible parameters for testing across engines
- Builds toward potential future systems with genuine geometric capabilities
- **It is a translation layer**, not literal math control

Scope honesty: This is **systematic prompting**, not CAD. The payoff is **clarity and repeatability**, not pixel determinism.

This mapping shows that the math terms aren't abstract, they connect back to measurable properties used inside the **Lens Structural Index (LSI)** system (see website for more information).

- **Centroid drift** (Δx): Does compositional gravity hold or scatter?
- **Edge fragmentation** (pr): Will structural boundaries maintain integrity?
- **Orientation stability** (θ): Does form resist chaotic reorientation?
- **Structural thickness** (ds): Is there substantive mass or just surface detail?

Each transform operator corresponds to a measurable structural primitive in the Lens Structural Index (LSI). This gives every move or transform traceable, testable, and consistent back to a scoring system that can look at success, for validation.

→ the LSI is a theoretical scoring engine within the Lens system that instead of measuring resemblance to training data, it measures resilience under structural pressure. It is not a required usage to obtain results or judge, it is just a methodology to apply math → translation between coordinates and words → translation between engines → some form of metrics. It should be viewed as directional if used until validated.

Operator	Effect	LSI Primitive	Meaning	Plain-English

<code>extend_vector</code>	Pulls a limb or form along a vector	$\Delta x \uparrow$	Centroid drift (measured displacement)	<i>How far the center of the form shifts from its original anchor.</i>
<code>spiral / hinge</code>	Coils or bends around an axis	$p_r \uparrow$	Rupture / torsion proxy (structural strain)	<i>How much twisting or coiling stress is added into the structure.</i>
<code>arc / tilt</code>	Rotates or tilts around anchor point	$\theta \uparrow$	Angular deviation (tilt / rotational force)	<i>How far the form tilts or rotates off its neutral alignment.</i>
<code>layer / skeleton</code>	Adds stacking, layering, or depth	$d_s \uparrow$	Skeleton depth (layering / volumetric force)	<i>How much perceived depth or layering is built into the figure.</i>

This mapping shows why the math exists: it turns abstract distortions into measurable moves. Each operator nudges a specific LSI dimension, so even experimental transformations stay predictable and repeatable. Instead of guessing at “stretch” or “coil,” you can see exactly which primitive is being manipulated and how it contributes to the overall structural score. → *See appendix 3 for more information.*

What It's Not

This is not a precision control system. Current image generation works through learned pattern associations, not coordinate transformations. The mathematical terminology creates systematic exploration tools, not literal mathematical execution.

- **Pattern matching, not solvers.** Text→image engines align prompts to training priors. “Parabolic extension” lands as a *recognizable smooth arc*, not a guaranteed quadratic curve.
- **Parameters = nudges, not constraints.** “Turns: ~1” or “gradual stretch” biases the sampler toward that look; it can still drift.
- **Stochastic + versioned.** Seeds, denoising schedules, and model updates change outcomes. Yesterday’s prompt ≠ today’s exact pixels.
- **Locks are perceptual, not geometric.** “Keep thickness/topology” pushes shading, silhouette, and continuity cues so it *reads* like the limb itself—not a proof of conservation.
- **Acceptance beats exactness.** We judge success by **visible criteria** (one clear overlap; wrist stays full; camera unchanged), not by measuring a curve.

Research Value

The framework’s strength lies in creating methodology for an unexplored area:

- Systematic approach to purposeful anatomical distortion
- Reproducible techniques for complex spatial transformations
- Cross-platform consistency for difficult prompt challenges
- Foundation for future integration with mathematical AI systems
- Explicit control channels (pose/depth/edges) and field-guided warps,
- Differentiable editors and 3D proxies,
- Engine APIs that accept true constraints (e.g., preserve cross-section while applying a spiral field).

Validation Approach

Results are validated through practical testing across multiple AI systems rather than theoretical precision. The framework succeeds if it reliably produces intended distortions, regardless of whether the underlying mathematical constructs reflect actual model behavior.

Deformation Operator Playbook should be evaluated as an **experimental research methodology** for achieving controlled anatomical distortions in AI-generated images - a genuinely difficult problem with no established solutions. Its value lies in systematic exploration of uncharted territory, not in theoretical accuracy about how AI systems work internally.

A Translation Layer for AI Image Control

As the playbook is not a direct mathematical control system for AI image generation. Instead, it serves as a **structured translation layer** between different AI capabilities and architectures. Current AI systems fall into two categories with fundamentally different capabilities:

- **Mathematical AI** (GPT with tools, code execution): Can perform precise calculations and coordinate transformations
- **Visual AI** (current text-to-image models): Translate language to images through learned pattern associations, not mathematical operations

These systems speak different languages, creating a gap when you want systematic, reproducible visual transformations.

Deformation Operator Playbook as Interface Protocol

The playbook Transform DSL functions as an intermediate representation that:

1. **Provides structure** that mathematical AI can generate and validate
2. **Uses consistent vocabulary** that visual AI can parse more reliably than free-form descriptions
3. **Enables transferability** across different AI architectures and future systems
4. **Maintains reproducibility** through parameterized operations

How Translation Works

User Intent → GPT (mathematical reasoning) → Deformation Operator Playbook specification → Image Model → Visual output. For example:

- User: "Make the arm spiral gracefully"
- GPT: Generates structured domain specific language with appropriate turns, scale, continuity parameters
- Image model: Processes structured language for more consistent results
- Future systems: Could potentially execute the mathematical operations directly

What It's Not

The playbook does not provide direct geometric control over current image generation models. Text-to-image systems work through learned correlations between language and visual patterns, not coordinate mathematics. The mathematical terminology creates systematic prompting, not literal mathematical execution.

Value Proposition

The system's strength lies in **systematic transferability** rather than precision control:

- Converts intuitive requests into structured specifications
- Provides consistent vocabulary across different AI systems GPT → Sora/Gemini, for example
- Creates a pathway for future AI architectures with genuine mathematical capabilities
- Improves current results through better structured prompting

Limitations

- Current image models don't implement the mathematical operations described
- Results depend on pattern matching, not geometric calculation
- Complexity may not justify improvements over well-crafted natural language prompts
- The "Lens Structural Index" is framework-specific terminology rather than established computer vision metrics, it can provide structured analysis, but is not covered here

Bottom Line

Deformation Operator Playbook should be evaluated as a **translation protocol** and **systematic prompting framework**, not as a direct mathematical control system. Its value lies in bridging the gap between AI systems that can reason mathematically and those that generate images through learned associations.

Deformation Operator Playbook

Quick Start

This system is designed so a new operator can get results on the very first run. Follow this five-step recipe:

1. Style binder → Pick a stable visual context (e.g., *charcoal, soft diffused light*).
2. Anchors → Place 2–4 anchors per limb (elbow, wrist, shoulder).
3. Select → Define the target region (e.g., `arm_L`).
4. Transforms → Apply a single operation (vector extend, sine modulation, arc, etc.).
5. Viewfinder → Adjust crop/shift/zoom to center the distortion and suppress artifacts.

How it works (in one line).

Anchors → Select → Transforms → Constraints → Viewfinder: a compact spec that lets you design single moves or stack them, then frame the result without adding new deformation.

Quick start (what a new user does).

1. Pick a stable **style binder** (e.g., charcoal),
2. place **anchors** (elbow, wrist, shoulder),
3. **select** the target region,
4. apply one **transform** (e.g., parabolic, Fibonacci coil),
5. adjust the viewfinder to center the distortion and suppress artifacts if desired.

What's inside (operators 0–8).

- Two starters Arm Extension and Sine Modulation
- Segmented Coil
- Vector Extension
- Logarithmic Scaling
- Rotational Transform
- Fibonacci Coil
- Parabolic Extension
- Depth Tug
- Sine Modulation
- Viewfinder Shift
- Each is framed with safe defaults, locks (e.g., thickness/topology), and continuity targets (C2) so deformations read as the limb itself, not added objects.
- GPT image, Sora prompt example.

Layout

- **Title bar:** “Deformation Operator Playbook”
- **Two-column grid:**
 - **Left column:** Operator name + short description (e.g., “*Parabolic extension, graceful stretch along axis with elbow lock*”).
 - **Right column:** YAML snippet (compact, syntax-colored).
- **Footer:** quick legend (C1 = continuity hinge, C2 = curvature, locks = topology/thickness preservation, expect = maps to LSI values).

Deformation Operator Playbook capsule = VCF in code language (safe defaults)

Quick Start Capsule 1: Arm Extension

This does a **gentle arm extension** anchored at the elbow, with locks to preserve thickness and prevent breaks. The **viewfinder** shift and zoom help center the distortion and suppress ghosting. This is a boilerplate for a first try.




```
vcf_capsule: 1.0
style: "charcoal, soft diffused light"

anchors:
  left_elbow: {x: 0.32, y: 0.58}
  wrist_L:    {x: 0.36, y: 0.78}

select:
  arm_L: {type: semantic, tag: "left_arm_forearm_hand"}

transforms:
- op: pin
  target: [left_elbow, wrist_L]
  stiffness: 0.95

- op: extend_vector
  target: arm_L
  anchor: left_elbow
  vec: {dx: 0.65, dy: 0.18}
  factor: 1.6
  locks: [thickness, topology]
  falloff: {type: gaussian, radius: 0.12}
  continuity: C1
  interpret: "single continuous arm; no branching"

constraints:
  seamless_integration: true
  material_consistency: "charcoal"

viewfinder:
  shift: {dx: -0.06, dy: +0.02}
  zoom: 1.03
```

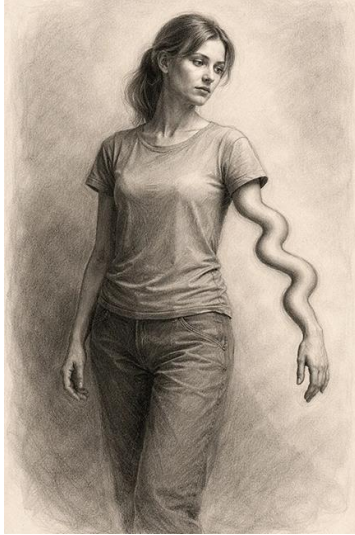
One-sentence prompt version (Quick Start 1: Arm Extension)

Charcoal, soft diffused light. A clothed female figure in a relaxed, neutral pose. Extend the left arm outward from the elbow along a vector angled slightly forward and right by a factor of about 2.6× its natural length. Preserve thickness and topology, with smooth anatomical continuity (C1) through the joints. Pin elbow and wrist to keep orientation stable; ensure a single continuous arm with no branching. Integrate seamlessly into the figure, consistent with charcoal texture and shading.

“Most common fixes”

- **Bend?** Use `hinge_bend` (22–30°) first, then a small `perspective_inverse_square` (0.10–0.15).
- **Coil looks like stretch?** Add `wrist_orient: pronate 8–12°` + `contact_occlusion` at overlaps.
- **Sine spawned extra arm?** Lower `amplitude` ≤ 0.012 , keep pins, keep `interpret`: clause.
- **Neck arc reads as elongation?** Use `arc_bend` (32–36°, `radius` 0.22–0.26); *optionally* a tiny `extend_parabolic` (≤ 0.18) to smooth.

Quick Start Capsule 2: Sine Modulation (Lite, Safe Defaults)



```
vcf_capsule: 1.0
style: "charcoal, soft diffused light"

anchors:
  left_elbow: {x: 0.33, y: 0.56}
  wrist_L:    {x: 0.38, y: 0.78}

select:
  arm_L: {type: semantic, tag: "left_arm_forearm_hand"}

transforms:
  - op: pin
    target: [left_elbow, wrist_L]
    stiffness: 0.96

  - op: modulate_sine
    target: arm_L
    axis: path
    amplitude: 0.020      # bumped for visible serpentine
    wavelength: 0.22      # longer beat to survive charcoal diffusion
    phase: 0.00
    locks: [topology, thickness]
    continuity: C2
    falloff: {type: gaussian, radius: 0.11}
    interpret: "single continuous arm; no branching"

  - op: wrist_orient
    target: hand_L
    mode: pronate
    angle_deg: 10
    continuity: C1

constraints:
  seamless_integration: true
  material_consistency: "charcoal"
  avoid: ["branching appendages", "prosthetic seams"]

viewfinder:
  shift: {dx: -0.04, dy: +0.02}
  zoom: 1.02
If you want the wave to read even stronger in the forearm, tighten the "frequency" locally:

lower wavelength to 0.18,
```

```
add segment_weight: {upper: 0.6, forearm: 1.0, hand: 0.9} in modulate_sine.
```

One-sentence prompt version (if they won't use YAML)

Charcoal, soft diffused light. Clothed figure in neutral pose. Apply a single continuous serpentine (sine) undulation left arm in proportion (not an extra limb): start at shoulder, end at the wrist. amplitude ≈ 0.012 , wavelength ≈ 0.20 , phase $\approx \pi/2$. Pin elbow and wrist so the hand remains intact; preserve thickness/topology; C2 continuity through joints; slight wrist pronation ($\sim 8^\circ$). No branching, no seams.

Tiny troubleshooting

- Spawns a “second arm”? → keep pins, include `interpret:"single continuous arm"`, lower amplitude to **0.010**.
- Wave too subtle? → nudge amplitude to **0.014** (keep pins on).
- Looks like stretch, not curve? → keep `wrist_orient` and add a light **contact occlusion** pass at the inner bend.

Things to know before deeper use:

FULL DISCLAIMER: Results vary by engine, model version, settings, and randomness. These operators describe intent, not guaranteed geometry, so expect iteration. Many platforms treat strong body deformations as low-quality “collapse” or safety risk; vendors routinely retune models to suppress such prompts, so behavior can change without notice. For the fastest path, work interactively with a conversational model: upload this Playbook, state the target engine (e.g., Sora), and ask for platform-specific rewrites or adjustments. Track versions/seeds and export accepted results promptly. Use only material you have rights to and follow each platform’s terms and content policies. No affiliation or endorsement is implied.

Selection & Anchors Primer

Anchors are the fixed points that give your transforms stability. Think of them as the “nails” that hold a limb, joint, or axis in place while the rest of the form is manipulated.

- **How many?**
Use 2–4 anchors per limb. At minimum: shoulder, elbow, and wrist for an arm; hip, knee, and ankle for a leg.
- **What if ignored?**
If the model skips your anchor, increase specificity (e.g., “left inner elbow crease” instead of just “elbow”) or reinforce with a `pin` operator.

Pin vs. Lock (cheat sheet):

- `pin = pose hold` → Keeps the anchor *stuck* in place relative to the canvas.
- `locks = geometry preservation` → Maintains proportions and thickness even under heavy transformation.

Operator Guardrails

Every operator has safe ranges and best practices. Use these to avoid runaway distortion.

- **Sine modulation:** Always pin **elbow + wrist** together, or the arm risks vanishing. Start with **amplitude ≤ 0.012** ; this keeps the serpentine subtle and anatomical. Increase slowly once stability is proven.
- **Perspective tug:** Always localize with a **falloff radius**. If you don’t, the entire figure may warp toward the vanishing point. To guarantee a bend, **add a pin on the upper arm**; this forces the tug to express as a local distortion rather than a global collapse.
- **Spiral/coil:** Beginners should avoid pure Fibonacci spirals. Instead, use a **segmented coil**: combine a joint rotation, a small wrist orientation, and a short spiral. This breaks the move into pieces the model can manage, producing smoother continuity without losing control.

Troubleshooting Table

When things go wrong, it usually falls into a few predictable categories. Here’s how to fix them:

- **Hand vanished?** Pin the wrist and add `locks:[topology, thickness]`. This prevents the model from deleting the hand when under strain.
- **Third arm appears?** Add `interpret:"single continuous arm; no branching"`. This tells the system to treat the limb as *one path only*.
- **Elongates instead of bending?** Swap the field operator for a **hinge/arc** operation first. Once a bend is established, you can add a smaller field on top for more nuance.

Engine-Agnostic Switches

Different models handle transforms differently. These switches help keep results consistent across engines:

- **Style dilution** If the chosen art style (charcoal, oil, digital) weakens your transform, increase **continuity** (C2) and reduce the global **magnitude**. This preserves the move without letting the style override it.
- **Seam appearance** If seams or breaks appear, add:
`falloff:{gaussian, radius:0.10-0.14}`
and `material_consistency:"charcoal"`.
This ensures shading stays smooth and the medium ties everything together.

LSI-Lite Mapping Stub

This table helps see why the math is in the operators. Each operation corresponds to a structural primitive:

- `extend_vector` → $\Delta x \uparrow$ (measured centroid drift)
- `spiral/hinge` → $\rho_r \uparrow$ (rupture / torsion proxy)
- `arc/tilt` → $\theta \uparrow$ (angular deviation)
- `layer/skeleton ops` → $d_s \uparrow$ (skeleton depth / layering)

VCF Transform DSL (capsule)

Conventions

- Coordinates: normalized `[0..1]` in image space (x=left→right, y=top→bottom).
- Angles: degrees. Lengths: fraction of image diagonal unless noted.
- `falloff`: how the transform blends to untouched areas (Gaussian | linear, radius).
- `continuity`: curvature continuity at seams (C0 pos, C1 tangent, C2 curvature).
- `locks`: properties preserved (area, thickness, proportion, topology).
- `expect`: intended LSI-lite shift (for sanity checking).

Deformation Operator Playbook (VCF) Transform DSL (capsule)

```
vcf_capsule: 1.0
anchors:
  left_elbow: {x: 0.32, y: 0.58}
  shoulder_L: {x: 0.30, y: 0.40}
  neck_base: {x: 0.50, y: 0.42}
  crop_center: {x: 0.46, y: 0.50}

select:
  arm_L: {type: "semantic", tag: "left_arm_forearm_hand"} # or polygon
  neck: {type: "semantic", tag: "neck"}
  field: {type: "rect", x: 0.00, y: 0.35, w: 1.00, h: 0.65}

transforms:
- op: extend_vector
  target: arm_L
  anchor: left_elbow
  vec: {dx: 0.65, dy: 0.18} # direction in norm coords
  factor: 1.8 # length scale along vec
  locks: [thickness] # keep arm width consistent
  falloff: {type: gaussian, radius: 0.12}
  continuity: C1
  expect: {Δx:+0.06, pr:+0.04}

- op: scale_log
  target: neck
  anchor: neck_base
  k: 0.75 # log scaling strength
  axis: y # elongate along vertical
  locks: [proportions_x]
  continuity: C2
  expect: {θ:+12, d_s:+0.02}

- op: rotate
  target: arm_L
```

```

    anchor: left_elbow
    angle: 30

- op: spiral_fibonacci
  target: arm_L
  center: shoulder_L
  turns: 1.25
  scale: 0.22
  handedness: CW
  blend: 0.6
  continuity: C1
  expect: {pr:+0.07}

- op: extend_parabolic
  target: arm_L
  axis: {dx: 1, dy: 0}
  factor: 0.35 # curvature magnitude
  vertex_lock: left_elbow
  continuity: C1

- op: modulate_sine
  target: arm_L
  axis: path # along centerline
  amplitude: 0.015
  wavelength: 0.18
  phase: 0.0
  continuity: C2
  expect: {K:+0.03}

viewfinder:
  shift: {dx: -0.10, dy: +0.04} # pre-render crop shift - Ghost Density basin
  zoom: 1.08

constraints:
  seamless_integration: true # no seams at body/coil junction
  material_consistency: "charcoal" # unify grain/lighting
  avoid: ["monstrous_enlargement"]

notes: "Graceful distortion; body neutral, distortion carries consequence."

```

Below is a tiny, Deformation Operator Playbook (capsule-ready) that lets you specify anchors, vectors, and exact transforms. It's readable, copy/pastable into prompts, and maps to Lens logic (Δx , r_v , p_r , θ , d_s) so you can predict consequence.

This example binder works for many of these, adjustments apparent in each capsule:

Binder:

Charcoal drawing, soft diffused light, short-sleeved figure on textured paper, neutral pose. The body itself deforms; keep continuous skin and believable volume; no ropes, bands, armor, or extra limbs; no text or watermarks.

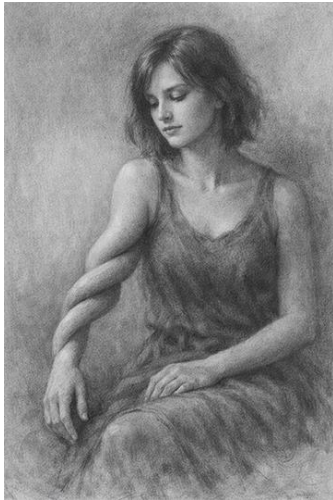
A charcoal drawing, soft diffused light. A female clothed figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joints.

INSERT DISTORTION

Maintain original thickness across the figure. Keep anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Disclaimer: Outputs may vary and engines change, some treat these prompts as "collapse." Iteration is expected. Use with your own assets and within platform policies. No guarantees or endorsements.

0) Segmented Coil (worked better than raw Fibonacci) * Recommended Starter Path



Segmented Coil - Standard

```
- op: segmented_coil
  target: arm_L
  elbow_rotate: 28
  wrist_pronate: 8
  spiral_turns: 0.85      # ' from 0.6 for clearer torsion rhythm
  locks: [thickness, topology]
  continuity: C2          # ' from C2/C1 mix to full C2
  shading: overlap_occlusion
```

Segmented Coil - Minimal (bare / short sleeve)

```
- op: segmented_coil Minimal (bare/rolled sleeve)
  target: arm_L
  elbow_rotate: 30
  wrist_pronate: 12
  spiral_turns: 0.90      # ' from 0.75 to ensure one visible crossing
  locks: [thickness, topology]
  continuity: C2
  shading: overlap_occlusion
```

Full capsule block (bare arm shown; keep garment section only if a cuff/roll is visible)

Alternatives: vcf_capsule: 0.3

style: "charcoal, soft diffused light"

layers:

```
  skin: {mask: "left_arm_forearm_hand"}
  garment:{mask: "left_sleeve_short"} # omit this whole layer for fully bare arm
```

anchors:

```
  shoulder_L: {x: 0.30, y: 0.40}
  elbow_L:    {x: 0.33, y: 0.58}
  wrist_L:    {x: 0.37, y: 0.78}
```

select:

```
  upper_arm_L: {type: semantic, tag: "left_upper_arm"}
  forearm_L:   {type: semantic, tag: "left_forearm"}
  hand_L:      {type: semantic, tag: "left_hand"}
  sleeve_L:    {type: semantic, tag: "left_sleeve_short"} # omit if bare
```

--- 1) Deform anatomy (skin layer) ---

transforms:

```
- op: pin
  target: shoulder_L
```

```

    stiffness: 0.95

- op: joint_rotate
  target: forearm_L
  hinge: elbow_L
  angle_deg: 30
  continuity: C2
  locks: [thickness, topology]

- op: wrist_orient
  target: hand_L
  mode: pronate
  angle_deg: 12
  continuity: C2

- op: spiral_segment
  target: forearm_L, hand_L
  center: elbow_L
  turns: 0.90          # ensures ≥1 clear overlap
  scale: 0.14          # : from 0.20 to avoid sausage/bulbous
  handedness: CW
  blend: 0.92          # smoother energy through elbow
  falloff: {type: gaussian, radius: 0.14} # : to distribute torsion along forearm
  continuity: C2
  locks: [thickness, topology]
  segment_weight: {forearm_proximal: 0.6, forearm_distal: 1.0, hand: 0.9} # new: elbow taper

- op: self_occlusion_enhance
  target: forearm_L
  style: "charcoal"
  params: {strength: 0.72, width: 0.010} # stronger crossing read

# --- 2) Conform garment with seam guards (only if sleeve visible) ---
- op: seam_lock
  target: sleeve_L
  seams: ["armhole", "cuff"]
  tolerance: 0.12

- op: garment_follow
  target: sleeve_L
  follow_target: forearm_L
  adherence: 0.55
  slip_bias: 0.25
  wrinkle_gen:
    mode: compression
    intensity: 0.50
    direction: "around coil"
    density: 0.55

constraints:
  seamless_integration: true
  material_consistency: "charcoal"
  interpret: "single continuous arm; no branching"
  avoid: ["rope", "cord", "wrapping", "external object"]
  anatomical_continuity: true
  require_overlap_crossing: true

viewfinder:
  shift: {dx: -0.05, dy: +0.02}
  zoom: 1.03

Net effect: more spiral (less sausage), smoother elbow handoff, clear self-crossing, and no "rope" read.
If you want, I can fire one render with these exact parameters (bare-arm variant) to confirm it lands.

```

Natural prompt A charcoal drawing, soft diffused light. A female clothed figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joins. Make the left arm itself twist into a gentle anatomical coil starting at the elbow, with one clear overlap that casts a soft occlusion shadow; keep the arm's thickness and the hand intact. Maintain original thickness across the figure. Keep anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Plain snippet: Make the **left arm itself** twist into a gentle coil starting at the **elbow**, with **one clear overlap** that casts a soft occlusion shadow; keep thickness and the hand intact.

Acceptance checks: arm mass (not an object) forms a coil; one visible overlap; thickness preserved; no seams/props.

Common failure → fix:

- Rope/bracelet shows up → add: *"the coil is the arm itself; not wrapping; no objects."*
- "Sausage" bulges → say: *"lower amplitude; slightly more turns; smooth, continuous curve through the elbow."*
- Elbow break → add: *"smooth transition at the elbow; no sharp corner."*

1) Vector Extension — *elbow anchored lengthening*



```
- op: extend_vector
  target: arm_L
  anchor: left_elbow
  vec: {dx: 0.65, dy: 0.18}
  factor: 1.8
  locks: [thickness]
  continuity: C1
```

Natural prompt: A charcoal drawing, soft diffused light. A female clothed figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joins. Lengthen the left arm outward from the elbow along a slight forward-right diagonal; almost double the natural length, but keep the forearm and hand the same thickness and fully connected. Maintain original thickness across the figure. Keep anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Plain snippet: Lengthen the **left arm outward from the elbow** along a slight forward-right diagonal—**nearly double** the natural reach—while **keeping the forearm/hand the same thickness** and fully connected.

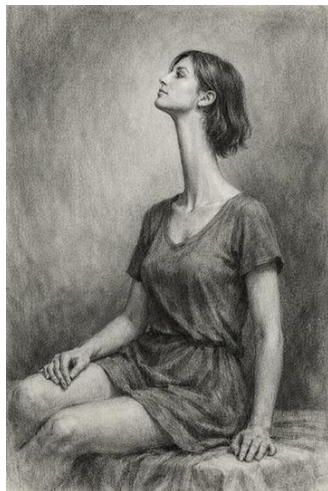
Acceptance checks: elbow stays put; direction is clear; hand size intact; no tearing at sleeve/skin.

Common failure → fix:

- Hand stretches → add: *"keep hand size and volume unchanged."*
- Direction drifts → add: *"extend along a forward-right line from the elbow."*

- Looks rubbery → reduce amount (e.g., “**just under double**”) and add: “*no thinning.*”

2) Logarithmic Scaling (Neck) — *gradual vertical stretch*



```
- op: scale_log
  target: neck
  anchors: neck_base
  k: 0.75
  axis: y
  locks: [proportions_x]
  continuity: C2
```

Natural prompt: A charcoal drawing, soft diffused light. A female clothed figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joins. Stretch the neck upward gradually so it lengthens more near the collar and less near the head; keep its width proportional and the jawline and shoulders coherent; leave the rest of the body unchanged. Maintain original thickness across the figure. Keep anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Plain snippet: Stretch the **neck upward gradually**—more near the collar, less near the head—keeping its **width proportional**; jawline and shoulders stay coherent; the rest of the body is unchanged.

Acceptance checks: gradient stretch (not uniform); neck width believable; head/shoulders stable.

Common failure → fix:

- Pinched “giraffe” neck → add: “*keep neck width proportional; preserve throat fullness.*”
- Head shrinks → add: “*head size unchanged.*”
- Shoulders creep → add: “*collarbones and shoulders stay in place.*”

3) Rotational Transform — *forearm twist around elbow*



```
- op: rotate
  target: arm_L
  anchor: left_elbow
  angle: 30
  continuity: C1
```

Natural prompt: A charcoal drawing on textured paper, soft diffused light. A young woman in a plain sweater stands in a neutral, relaxed pose. Her left arm extends outward, stretched noticeably longer than normal, about one and a half times its usual length. The extension originates cleanly at the elbow and continues in a smooth, natural line toward the lower right of the frame. The forearm and hand remain thin and proportional, without swelling or distortion, as though gracefully elongated. The sweater sleeve follows the stretch, preserving fabric folds and shading, with charcoal grain and tonal shading consistent across the entire figure. The overall style is realistic yet expressive, emphasizing the tension between the contained body and the extended limb.

Natural prompt #2: A charcoal drawing, soft diffused light. A female clothed figure in a neutral, relaxed pose. Seamless integration between body and any distortion. Her left arm extends straight outward from the shoulder, stretched noticeably longer than normal, about one and a half times its usual length. The arm turns downward at the elbow and continues in a smooth, natural line toward the lower right of the frame. The forearm and hand remain thin and proportional, without swelling or distortion, as though gracefully elongated. Maintain original thickness across the figure. Keep anatomy and proportions elsewhere. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Plain snippet: Freeze the **shoulder/upper arm**. Keep the **elbow angle the same**. Rotate the **forearm in place around the elbow** so the **palm turns toward the torso** and the **thumb rolls downward**; add soft spiral skin folds near the elbow and a few sleeve wrinkles that **lag the twist**.

Acceptance checks: upper arm unmoved; elbow angle unchanged; twist reads at forearm; thickness preserved.

Common failure → fix:

- Looks like a whole-arm pose → add: *“upper arm stays fixed; rotation happens **only** at the forearm.”*
- Reads as bend, not twist → add: *“no extra bend; palm turns inward; subtle spiral crease elbow→wrist.”*
- Wrist collapses → add: *“keep wrist full; no thinning.”*

4) Fibonacci Coil — *tight-proximal, open-distal spiral*



```
- op: spiral_fibonacci
  target: arm_L
  center: shoulder_L
  turns: 1.1          # ' From 1.25, avoids over-wrapping and segment crowding
  scale: 0.16         # ' From 0.22, reduces "sausage" bulge
  handedness: CW
  phi_ratio: 1.618    # explicit golden-ratio growth (was implicit)
  blend: 0.90         # ' From 0.6, smoother shoulder - biceps integration
  falloff: {type: gaussian, radius: 0.18} # ' distributes torsion more evenly
  continuity: C2      # ' From C1, keeps curves smooth and anatomical
  locks: [thickness, topology]
  segment_weight: {proximal: 0.7, distal: 1.0} # new: distal opens wider, proximal tighter
  require_overlap_crossing: true
  avoid: ["rope", "cord", "external wrapping"]
  anatomical_continuity: true
```

Natural Prompt: A charcoal drawing, soft diffused light. A female clothed figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joins. Form a graceful spiral in the left arm that's tighter near the shoulder and opens wider toward the forearm and hand; show one visible overlap; the spiral is the arm itself, not something wrapped around it. Maintain original thickness across the figure. Keep anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

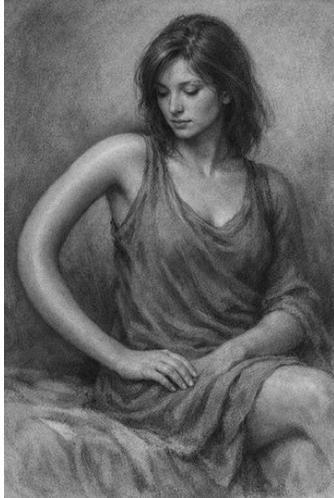
Plain snippet: Form a **graceful spiral** in the **left arm** that is **tighter near the shoulder** and **opens wider toward the forearm and hand**; show **one visible overlap**; the spiral is the **arm itself**, not something wrapped around it.

Acceptance checks: spacing tighter at origin, wider distally; visible overlap; continuous arm mass; no props.

Common failure → fix:

- Even "toroid" bulges → add: *"tighter near shoulder, wider toward hand."*
- Rope literalization → add negatives: *"not wrapping; no rope/band/bracelet."*
- Hard break at shoulder → add: *"smoothly blends into upper arm."*

5) Parabolic Extension — *graceful arc with elbow as crest*



```
- op: extend_parabolic
  target: arm_L
  axis: {mode: "from_to", from: shoulder_L, to: wrist_L}
  factor: 0.42
  vertex_lock: elbow_L
  continuity: C2
  locks: [thickness, topology]

  curvature_tension: 0.72
  falloff: {type: gaussian, radius: 0.14}
  distal_taper: 0.80
  proximal_weight: 0.70
  asymmetry_bias: 0.20

  self_occlusion_enhance:
    strength: 0.70
    width: 0.012
    side: "inner_curve"

  anatomical_continuity: true
  require_elbow_continuity: true
  avoid: ["ballooning", "crease seam"]
```

B) Grace Variant (softer arc for "elegant stretch")

yaml

Copy code

```
- op: extend_parabolic
  target: arm_L
  axis: {mode: "from_to", from: shoulder_L, to: wrist_L}
  factor: 0.36
  vertex_lock: elbow_L
  continuity: C2
  locks: [thickness, topology]

  curvature_tension: 0.68
  falloff: {type: gaussian, radius: 0.19}
  distal_taper: 0.84
  proximal_weight: 0.74
  asymmetry_bias: 0.18

  self_occlusion_enhance:
    strength: 0.62
    width: 0.010
    side: "inner_curve"

  anatomical_continuity: true
```

```

require_elbow_continuity: true
avoid: ["ballooning", "crease seam"]

```

Natural Prompt: A charcoal drawing, soft diffused light. A female clothed figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joints. Shape the left arm into a smooth, graceful U shape arc with the elbow smoothed into a U basin (NO joint) as the high point. The curve runs from shoulder to wrist as one continuous noodle sweep. Keep the arm full—no sausage bulges, no hard crease. Add a faint shadow along the inner side of the arc to make the curve read. Maintain original thickness across the figure. Keep anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Plain snippet: Shape the **left arm** into a **smooth, elegant arc** with the **elbow as the high point**; a single continuous curve shoulder→wrist; avoid bulging or a sharp crease; add a **faint inner-arc shadow** to make the curve read.

Acceptance checks: elbow clearly the crest; one continuous arc; no bulge/kink; volume preserved.

Common failure → fix:

- Reads straight → say: *"increase the arc; make elbow visibly the highest point."*
- Kink at elbow → add: *"no sharp corner; smooth handoff shoulder↔forearm."*
- Bulging → say: *"gentle arc; keep thickness even; avoid ballooning."*

6) Depth Tug — *localized pull (not a camera change)*



```

# Pre-stability (keeps the hinge clean)
- op: pin
  target: shoulder_L
  stiffness: 0.95
- op: hinge_lock
  target: elbow_L
  stiffness: 0.85

# Depth pull
- op: perspective_inverse_square
  target: arm_L
  center: forearm_mid_L
  vanishing: {x: 0.85, y: 0.42}
  magnitude: 0.44
  continuity: C2
  locks: [topology]
  thickness_lock: 0.90
  silhouette_preserve: 0.86

# localizes the field
# keep your VP
# ! for readable convergence
# ! from C1 to avoid kinks
# prevents rubber-hose thinning

```

```

falloff:
  mode: inverse_square
  radius: 0.16 # tighter, less bleed into upper arm
  axial_bias: {along: 0.70, across: 0.30}
  clamp: 0.90

z_bias: 0.18 # forward pop toward VP
torsion_compensation: 0.20 # suppress unintended twist
distal_scale_comp: 0.92 # protects wrist/hand fullness

wrist_plane_adjust:
  mode: align_to_vp
  angle_deg: 6 # subtle alignment cue

depth_occlusion_enhance:
  strength: 0.68
  width: 0.010
  region: "under-forearm_to_wrist"

anatomical_continuity: true
require_vanishing_convergence: true
convergence_tolerance: 0.05
avoid: ["accordion_fold", "thin_wrist_collapse", "edge chatter"]

# Optional viewfinder bias to make the VP read harder (safe)
viewfinder:
  shift: {dx: +0.02, dy: 0.00}
  zoom: 1.02

If you want a softer version for conservative cases, set magnitude: 0.40, radius: 0.18, z_bias: 0.14.

```

Natural Prompt: Charcoal female study on textured paper. Do not change the camera. Modify only the left arm and pull strongly forward to viewer, take forearm (elbow→hand): pull it toward a small point just to the right of the frame so the top and bottom edges subtly converge, strongest near the wrist. Upper arm and elbow stay unchanged. Keep wrist/hand full (no thinning). Add tighter wrist-crease spacing, a narrow soft shadow under the forearm, and a slightly brighter near edge to sell forward depth.

Plain snippet: Do not change the camera. Modify **only the left forearm (elbow→hand)**: make both **top and bottom edges subtly converge** toward a small point just to the **right** of the frame, **strongest near the wrist**; the **upper arm stays unchanged**. Keep the wrist/hand full; add **tighter wrist-crease spacing**, a **narrow soft under-forearm shadow**, and a **slight highlight on the near edge** to sell the forward pop.

Acceptance checks: convergence strongest at wrist; upper arm untouched; wrist/knuckles show micro-compression; slim under-shadow present.

Common failure → fix:

- Looks like generic foreshortening → add: *"upper arm frozen; effect only from elbow to hand; camera unchanged."*
- No depth read → add: *"denser wrist creases and slight knuckle compression; narrow under-forearm shadow."*
- Wrist thins → add: *"keep wrist full; no thin-wrist collapse."*

7) Sine Modulation — *serpentine wave*



```
- op: modulate_sine
  target: arm_L
  axis: path
  amplitude: 0.012      # tuned; 0.010-0.016 safe
  wavelength: 0.18
  phase: 1.57           #  $\pi/2$ , lateral bow start
  locks: [thickness, topology]
  continuity: C2
  falloff: {type: gaussian, radius: 0.10}
  interpret: "single continuous arm; no branching"
  expect: {pr:+0.05}
```

Natural Prompt: A charcoal drawing of a clothed female figure in a neutral, relaxed pose. The figure's left arm is a thin distinct, sculpted coil that spirals away from the body with the hand at the end. The arm is a piece of flexible form, with a taut, skin surface and an internal, spiraling torsion that does not enlarge. This coiled form is then placed onto the figure, maintaining its connection to the shoulder and hand. The entire composition should be a single, seamless image, with no visible seams or joints, where the organic body meets the spiraling, un-resolvable form.

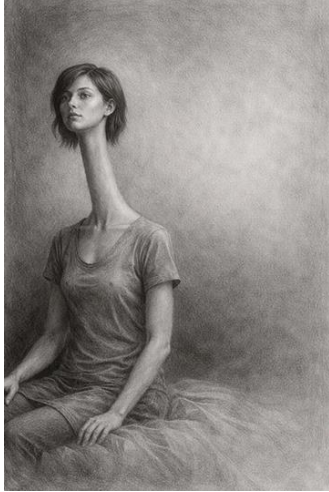
Plain snippet: Add **one gentle serpentine wave** running from **elbow to hand**; joints feel pinned and continuous; the **limb mass follows the wave** (not just surface wrinkles); hand remains intact.

Acceptance checks: single crest/trough across the forearm; joints connected; volume intact.

Common failure → fix:

- Surface wrinkle only → add: *"the limb's mass follows the wave; not just skin."*
- Over-wobble → *"smaller wave; one gentle undulation only."*
- Wrist collapse → *"keep wrist/hand full."*

8) Viewfinder Shift "Ghost Density" (pre-render crop shift)



```
viewfinder:
  shift: {dx: -0.12, dy: +0.08}
  zoom: 1.05
```

Word prompt: Do not change anatomy. Reframe the same figure by pushing the subject toward the left edge and leaving a wide, airy void on the right, with a slight zoom to raise edge tension

Or with deformation

A) Base frame (carry your approved QS-5 exemplar)



```
# QS-5 - Parabolic Extension (exemplar v0.8)
- op: extend_parabolic
  target: arm_L
  axis: {mode: "from_to", from: shoulder_L, to: wrist_L}
  factor: 0.42
  vertex_lock: elbow_L
  continuity: C2
  locks: [thickness, topology]
  curvature_tension: 0.72
  falloff: {type: gaussian, radius: 0.14}
  distal_taper: 0.80
  proximal_weight: 0.70
  asymmetry_bias: 0.20
  self_occlusion_enhance: {strength: 0.70, width: 0.012, side: "inner_curve"}
```

```

anatomical_continuity: true
require_elbow_continuity: true
avoid: ["ballooning", "crease seam"]

```

Word prompt: Generate a second shot of the same scene with identical pose and geometry; change only the framing: push the subject left, keep a large right-side negative space, slight zoom; no anatomical changes.

B) Viewfinder overlay for QS-8 (apply after QS-5, no new deformation)



```

# QS-8 - Viewfinder (ghost density)
viewfinder:
  shift: {dx: -0.20, dy: +0.06} # pushes subject toward left edge
  zoom: 1.10 # enlarges to raise edge pressure
  edge_proximity: {left: 0.85} # shoulder/ear near frame (tension)
  void_bias: {right: 1.25} # preserve a large right-side void
  crop_allowance: {shoulder: 0.02, forearm: 0.03}

constraints:
  interpret: "no geometric deformation added; framing only"
  material_consistency: "charcoal"
  preserve_transforms: ["extend_parabolic"] # **do not** alter the QS-5 geometry

```

Natural prompt: Generate a second shot of the **same scene with identical pose and geometry**; change only the framing: push the subject left, keep a large right-side negative space, slight zoom; no anatomical changes.

Plain snippet: **Do not change anatomy.** Re-frame the same figure by pushing the subject toward the **left edge** and leaving a **wide right-side void**, with a **subtle zoom** to raise edge tension; the pose and body remain identical.

Acceptance checks: geometry identical; left-edge proximity; big right-side negative space; tension rises.

Common Failure → Fix:

- Geometry changed → restate: *"framing only; pose and body unchanged."*
- Effect too mild → *"increase shift/zoom slightly."*
- Harsh crop → *"allow a small trim on shoulder/forearm; keep face unobstructed."*

Authorship

This framework was architected by Russell Parrish and recursively co-developed inside GPT, Gemini and Claude. Every critique is human-led; every recursion is model-driven. The result: a reasoning layer authored through language, not image manipulation.

This isn't a theory. It's already running.

If you're building generative tools, or trying to make them think better, this is your bridge.

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All non-code materials in this project, including the system design, frameworks (e.g., Visual Lens, Sketcher, Critique Room Protocol, Deformation Operator Playbook),

explanatory text, diagrams, and all visual outputs are protected as original intellectual property and may **not** be copied, reproduced, distributed, or transmitted in any form including inclusion in AI training datasets, without prior written permission from the creator.

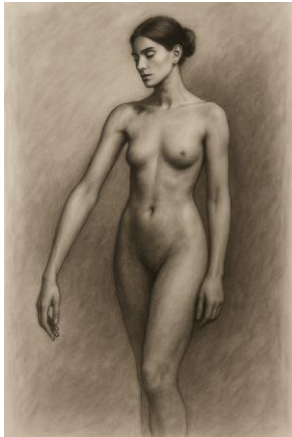
Code & DSL snippets license (MIT).

All source-like examples (code blocks, DSL snippets, JSON/YAML operator specs) are released under the MIT License in the appendix. You may use, copy, modify, merge, publish, distribute, sublicense, and/or sell derivative works of those **snippets only**, subject to the MIT terms. This permissive grant **does not** apply to the non-code materials above.

Appendix 1

Case Studies → GPT/Sora/Gemini

GPT: Case Study 1



Natural Prompt

A charcoal drawing, soft diffused light. A female figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joins. The left arm extends precisely along a vector originating at the left elbow, oriented toward the lower-right ($\approx 0.65, 0.18$ direction) and lengthened by a factor of **1.8×** while **maintaining original thickness** across the forearm and hand. Keep curvature continuity at the elbow (**C1** tangent continuity), preserve anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Structured capsule (VCF Transform DSL)

```
vcf_capsule: 1.0
style: "charcoal, soft diffused light"
anchors:
  # Approx normalized coords; adjust if you prefer
  left_elbow: {x: 0.32, y: 0.58}

select:
  arm_L: {type: "semantic", tag: "left_arm_forearm_hand"}

transforms:
  - op: extend_vector
    target: arm_L
    anchor: left_elbow
    vec: {dx: 0.65, dy: 0.18}  # direction (toward lower-right)
    factor: 1.8                # 1.8× extension along that vector
    locks: [thickness]        # preserve cross-sectional width
    falloff: {type: gaussian, radius: 0.12}
    continuity: C1             # tangent continuity at the elbow
    expect: {Δx:+0.05, pr:+0.02}

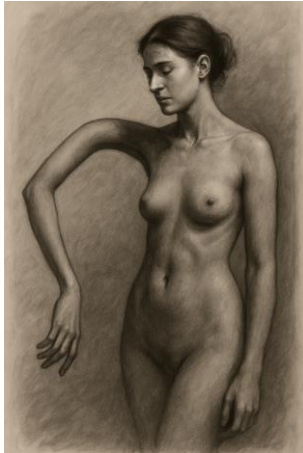
constraints:
  seamless_integration: true
  material_consistency: "charcoal"
  avoid: ["monstrous_enlargement", "prosthetic_seams"]
```


notes: "Neutral body; distortion carries consequence—graceful, not grotesque."

If you want a tiny boost toward off-center tension (Ghost Density basin), add this (optional):

```
viewfinder: {shift: {dx:-0.08, dy:+0.03}, zoom: 1.05}
```

Case Study 2



A) Stronger, localized field bend (keep topology, bias the forearm)

Natural prompt

A charcoal drawing, soft diffused light. A female figure in a neutral, relaxed pose. Bend the **left forearm** toward a vanishing point at (0.85, 0.42) using an **inverse-square field** with **high local influence** around the elbow; shoulder remains stable. Preserve topology and thickness; maintain C2 curvature continuity through the elbow so the forearm **curves** toward the vanishing point rather than simply stretching. Keep seamless integration and consistent charcoal grain.

VCF capsule

```
vcf_capsule: 1.0
```

```
style: "charcoal, soft diffused light"
```

```
anchors:
```

```
  shoulder_L: {x: 0.30, y: 0.40}
```

```
  left_elbow: {x: 0.32, y: 0.58}
```

```
select:
```

```
  upper_arm_L: {type: semantic, tag: "left_upper_arm"}
```

```
  forearm_L: {type: semantic, tag: "left_forearm_hand"}
```

```
transforms:
```

```
  # Keep the upper arm mostly fixed so the bend localizes in the forearm
```

```
  - op: pin
```

```
    target: upper_arm_L
```

```
    stiffness: 0.9
```

```
  # Stronger, tighter field centered at the elbow region
```

```
  - op: perspective_inverse_square
```

```
    target: forearm_L
```

```
    vanishing: {x: 0.85, y: 0.42}
```

```
    magnitude: 0.85 # , from 0.4 to force curvature
```

```
    locks: [topology, thickness]
```

```
    falloff: {type: gaussian, radius: 0.08} # tighter radius - localized bend
```

```
    continuity: C2
```

```
    segment_weight: {proximal: 0.7, distal: 1.0} # bend distal more than near elbow
```

```
    expect: {theta: +12, delta_x: +0.05, delta_s: +0.02}
```

```
constraints:
  seamless_integration: true
  material_consistency: "charcoal"
  avoid: ["monstrous_enlargement", "prosthetic seams"]

notes: "Localize curvature in forearm; shoulder/upper arm remain stable."
```

B) Explicit hinge + target arc (deterministic bend)

If A still under-bends, use an explicit bend operator (then optionally blend a small field so it reads as perspective, not a mechanical joint).

Natural prompt

A charcoal drawing, soft diffused light. A female figure in a neutral pose. **Hinge-bend** the left forearm around the elbow with **C2 curvature** to form an arc subtending **22–28°** toward a virtual vanishing point at (0.85, 0.42). Preserve thickness and topology; maintain seamless integration and coherent charcoal grain. The result should **visibly curve** toward the right with depth bias, not stretch.

VCF capsule

```
vcf_capsule: 1.0
style: "charcoal, soft diffused light"

anchors:
  shoulder_L: {x: 0.30, y: 0.40}
  left_elbow: {x: 0.32, y: 0.58}

select:
  forearm_L: {type: semantic, tag: "left_forearm_hand"}

transforms:
- op: hinge_bend
  target: forearm_L
  hinge: left_elbow
  arc_degrees: 26          # set the bend amount
  radius: 0.22             # in image-diagonal units
  plane_bias: {toward: {x: 0.85, y: 0.42}, weight: 0.7}
  locks: [thickness, topology]
  continuity: C2

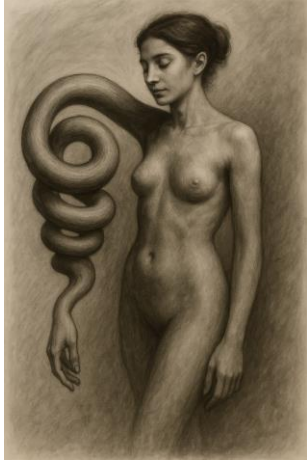
# tiny perspective field to add depth feel (optional)
- op: perspective_inverse_square
  target: forearm_L
  vanishing: {x: 0.85, y: 0.42}
  magnitude: 0.15
  falloff: {type: gaussian, radius: 0.10}

constraints:
  seamless_integration: true
  material_consistency: "charcoal"
```

Quick tuning levers

- **More bend:** increase magnitude (A) or arc_degrees (B).
- **Bend location:** shrink radius (A) or radius + plane_bias.weight (B) to keep it near the elbow.
- **Avoid stretch:** keep locks: [thickness, topology] and continuity: C2.
- **Make it read as depth, not 2D warp:** keep a small perspective_inverse_square after the hinge (B) or set segment_weight to bias distal curvature (A).

Case Study 3



Natural Prompt

A charcoal drawing, soft diffused light. A female figure in a neutral, relaxed pose. Enforce two geometric distortions with explicit joint constraints and shading cues:

1. **Left arm – segmented coil:** Forearm rotated around the elbow by $\sim 34^\circ$. Wrist oriented with slight pronation ($\sim 18^\circ$). Hand follows a clockwise spiral of ~ 0.85 turns, gentle growth, with clear foreshortening. Preserve thickness, topology, and seamless integration. Cast shadow and occlusion added so coils overlap visibly, with darkened charcoal shading at overlaps.
2. **Neck – hard arc:** Bend the neck upward in a clear 36° arc, radius ~ 0.22 , vertex locked at the base of the neck. Maintain proportional width, C2 continuity at jaw/clavicle. Head tilted slightly ($\sim 8^\circ$) to follow the arc plane. Add subtle charcoal shading cues (SCM, suprasternal notch) to emphasize anatomical arc.

Keep anatomy coherent elsewhere; no seams; material consistency: charcoal grain and soft diffused lighting. Slight off-center framing to heighten latent tension.

Settings:

- Size: 1024×1024
- n: 1
- transparent_background: false

```
transforms:
# ARM – enforce coil
- op: joint_rotate
  target: forearm_L
  hinge: left_elbow
  angle_deg: 34          # ' from 28 - visible bend
  continuity: C1
  locks: [thickness, topology]

- op: wrist_orient
  target: hand_L
  mode: pronate          # force wrist rotation so hand follows the coil
  angle_deg: 18
  continuity: C1

- op: spiral_segment
  target: hand_L
  center: wrist_L
  turns: 0.85            # ' from 0.6
  scale: 0.26            # modest growth
  handedness: CW
  blend: 0.90            # adhere to spiral path
  falloff: {type: gaussian, radius: 0.09}
  locks: [thickness, topology]
```

```

continuity: C1

- op: contact_occlusion
  target: [forearm_L, hand_L]
  style: "charcoal"
  params: {strength: 0.6, width: 0.012} # darken overlaps to read as coil

# NECK – explicit arc, then tiny smooth
- op: arc_bend
  target: neck
  hinge: neck_base
  arc_degrees: 36 # ↑ curvature
  radius: 0.22 # ↑ radius = clearer arc
  orientation: up
  continuity: C2
  locks: [topology, width]

- op: head_chin_tilt
  target: head
  pivot: jaw_pivot
  angle_deg: 8 # commit head to the arc plane

- op: anatomical_shading_hint
  target: neck
  cues: ["SCM", "suprasternal_notch"]
  style: "charcoal"
  strength: 0.4

# keep fields minimal so geometry isn't flattened
# viewfinder nudge to lean into Ghost Density
viewfinder: {shift: {dx: -0.06, dy: +0.02}, zoom: 1.04}

```

If you want to go even harder (nuclear option)

Replace spiral_segment with **two-step coil**:

```

- op: joint_rotate {target: hand_L, hinge: wrist_L, angle_deg: 22, continuity: C1}
- op: path_conform {target: hand_L, path: "fibonacci", turns: 0.6, scale: 0.24, adherence: 0.85}

```

That removes the model's chance to "straighten" the path.

Visual cues to add in the prose (helps the renderer)

- **"Foreshortened knuckles** overlap the forearm as the coil turns toward the viewer."
- **"Cast shadow** from the upper coil onto the lower segment."
- **"Chin follows the neck's arc,** slight lift toward the bend."

Case Study 4



Natural Prompt

A charcoal drawing, soft diffused light. A female figure in a neutral, relaxed pose. Apply a single continuous serpentine sine modulation along the existing left arm (not an extra limb). Elbow and wrist are anchored, keeping the hand intact. Gentle undulation: amplitude ~ 0.012 , wavelength ~ 0.20 , phase offset $\sim \pi/2$ for a lateral bow. Preserve thickness, topology, and seamless continuity. Add a slight wrist pronation ($\sim 8^\circ$) so foreshortening emphasizes the 3D wave. No duplication, no branching, no seams. Consistent charcoal grain and diffused light.

VCF Transform DSL (v2.1)

```
vcf_capsule: 2.1
style: "charcoal, soft diffused light"

anchors:
  left_elbow: {x: 0.32, y: 0.58}
  wrist_L:    {x: 0.36, y: 0.78}

select:
  arm_L: {type: semantic, tag: "left_arm_forearm_hand"}

transforms:
  - op: pin
    target: [left_elbow, wrist_L]
    stiffness: 0.95

  - op: modulate_sine
    target: arm_L
    axis: path
    amplitude: 0.012
    wavelength: 0.20
    phase: 1.57          #  $\pi/2$ , lateral bow
    locks: [thickness, topology]
    continuity: C2
    falloff: {type: gaussian, radius: 0.10}
    interpret: "single continuous arm; no branching"
    expect: {pr:+0.05}

  - op: wrist_orient
    target: hand_L
    mode: pronate
    angle_deg: 8
    continuity: C1

constraints:
  seamless_integration: true
  material_consistency: "charcoal"
  avoid: ["branching appendages", "prosthetic seams"]
```

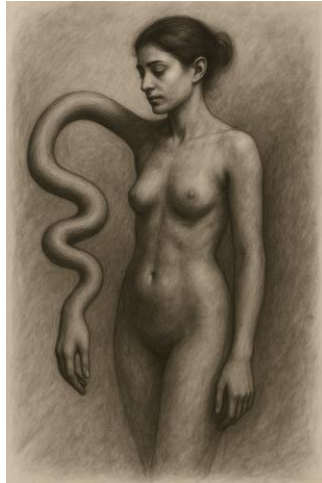
```

viewfinder:
  shift: {dx: -0.05, dy: +0.02}
  zoom: 1.03

```

This combo **pins + sine + small wrist pronation** — is what kept the arm continuous and prevented the model from spawning a duplicate.

Case Study 5



Natural prompt

A charcoal drawing, soft diffused light. A female figure in a neutral, relaxed pose. Apply a single continuous serpentine modulation along the existing left arm (not an extra limb). Elbow and wrist are pinned so the hand remains intact. Undulation parameters: amplitude ≈ 0.016 , wavelength ≈ 0.16 , phase $\approx \pi/2$ to begin with a lateral bow. Preserve thickness and topology; maintain C2 continuity through elbow and wrist. Add a slight wrist pronation ($\sim 10^\circ$) so foreshortening sells the 3D wave. No duplication, no branching, no seams. Consistent charcoal grain and soft diffused light.

VCF Transform DSL (v3)

```

vcf_capsule: 3.0
style: "charcoal, soft diffused light"

anchors:
  left_elbow: {x: 0.32, y: 0.58}
  wrist_L:    {x: 0.36, y: 0.78}

select:
  arm_L: {type: semantic, tag: "left_arm_forearm_hand"}

transforms:
  - op: pin
    target: [left_elbow, wrist_L]
    stiffness: 0.96

  - op: modulate_sine
    target: arm_L
    axis: path
    amplitude: 0.016      # 1 for clearer serpentine
    wavelength: 0.16      # tighter frequency
    phase: 1.57           #  $\pi/2$ 
    locks: [thickness, topology]
    continuity: C2
    falloff: {type: gaussian, radius: 0.10}
    interpret: "single continuous arm; no branching"
    expect: {pr:+0.06}

```

```

- op: wrist_orient
  target: hand_L
  mode: pronate
  angle_deg: 10
  continuity: C1

constraints:
  seamless_integration: true
  material_consistency: "charcoal"
  avoid: ["branching appendages", "prosthetic seams"]

viewfinder:
  shift: {dx: -0.05, dy: +0.02}
  zoom: 1.03

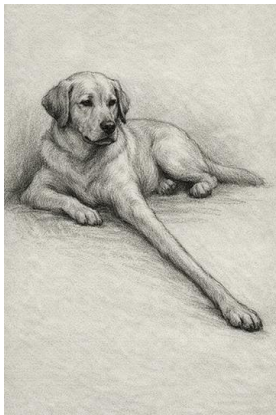
```

If you want an even bolder read without risking a “second arm,” nudge:

- amplitude: 0.018
- keep both pins on
- add a small **contact occlusion** pass at the inner wave turns to deepen the charcoal where curvature tightens.

Sora

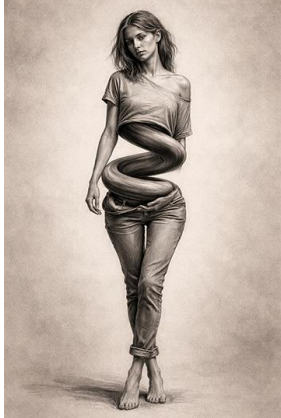
Case Study 6



Natural Prompt

A drawing, soft diffused light. A dog in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joins. The left legs extends precisely along a vector originating at the left elbow, oriented toward the lower-right ($\approx 0.65, 0.18$ direction) and lengthened by a factor of $1.8\times$ while maintaining original thickness across the forearm and hand. Keep curvature continuity at the knee, preserve anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Case Study 7



Natural Prompt

A charcoal drawing of a clothed female figure in a neutral, relaxed pose. The figure's torso is a distinct, sculpted coil that spirals linking the bottom half of the body with the top. The coil is a piece of flexible form, with a taut, smooth surface and an internal, spiraling torsion. This coiled form is then placed onto the figure, maintaining its connection to the chest and hip. The entire composition should be a single, seamless image, with no visible seams or joints, where the organic body meets the spiraling, un-resolvable form.

Case Study 8



A charcoal drawing of a figure, posed with head gently tilted down. Left arm begins normally at the shoulder but then twists into a sculptural coil, spiraling outward in a continuous loop before descending into a hand. The coil feels taut and continues skin, with an internal torsion that suggests controlled distortion rather than collapse. The rest of the body remains calm and still, contrasting with the coiled, impossible limb. Charcoal rendering with soft, diffused light and atmospheric shading.

Case Study 8 (has math, but Sora largely ignores, just used as a test)



Natural Prompt

A painting of a woman in a neutral, relaxed pose. The figure's left arm is a distinct, sculpted coil that spirals away from the body with the hand at the end. The arm is a piece of flexible form, with a taut, smooth surface and an internal, spiraling torsion. This coiled form is then placed onto the figure, maintaining its connection to the shoulder and hand. The entire composition should be a single, seamless image, with no visible seams or joints, where the organic body meets the spiraling, un-resolvable form.

Case Study 9



Natural Prompt

A charcoal drawing, soft diffused light. A female figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joins. The left arm extends precisely along a vector originating at the left elbow, oriented toward the lower-right ($\approx 0.65, 0.18$ direction) and lengthened by a factor of $1.8\times$ while maintaining original thickness across the forearm and hand. Keep curvature continuity at the elbow, preserve anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

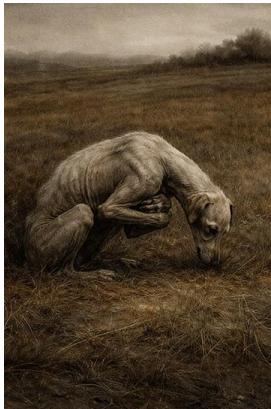
Case Study 10



Natural Prompt

A painting in the style of Andrew Wyeth. A lone female figure is in an old barn. The figure's body is contorted with grief and sorrow. Her head is bowed, and her neck appears painfully elongated, pulling her gaze downwards. Her arms are drawn in towards her body, perhaps also elongated and thin, emphasizing a sense of collapse and internal suffering. The overall mood is one of profound, desolate grief and physical strain. The color palette is muted, with heavier shadows, browns, grays, and earthy tones, enhancing the somber atmosphere.

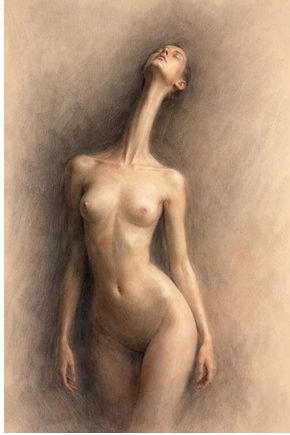
Case Study 11



Natural Prompt

A painting in the style of Andrew Wyeth's 'Christina's World.' A lone dog lies in a field of dry, tawny grass. The dog's body is a gaunt, skeletal form, hunched and pulled inward by an immense, unseen force. Dog's elongated neck and spine are painfully strained, forcing the dog's head to bend deeply towards the ground. Dog's arms are drawn in tightly, pressing against her chest in a gesture of profound suffering. The overall mood is one of desolate physical pain and internal collapse. The color palette is muted, with heavier shadows, browns, grays, and earthy tones.

Case Study 12



Natural Prompt

A pastel drawing of a female figure. Her body is in a neutral, relaxed pose, contrasting with her neck which is dramatically stretched towards the top of the frame. The elongated neck remains thin and in proportion to the rest of the body, creating a sense of graceful distortion rather than monstrous enlargement. The drawing uses soft, diffused lighting with subtle shadows, enhancing the charcoal texture and the figure's form without harsh lines. The overall style is expressive and slightly melancholic, highlighting the tension between the figure's contained body and the outstretched limb.

Case Study 13



Natural Prompt

A clothed man in shorts sitting in a neutral, relaxed pose on a couch. His legs are a distinct, sculpted coil that spirals away from the body with the feet at the end, which must be in proportion to the body. The legs are a piece of flexible form, with a taut, skin surface and an internal, spiraling torsion. This coiled form is then placed onto the figure's thighs, maintaining its connection to the hip and foot. The entire composition should be a single, seamless image, with no visible seams or joints, where the organic body meets the spiraling, un-resolvable form.

Case Study 14



Natural Prompt

A photograph of a cozy café interior. Warm afternoon light through large windows. Wooden tables, coffee cups, pastries on display. A man sits at a corner table, relaxed posture, casual sweater and jeans. **DEFORMATION - ARM EXTENSION:** His left arm extends from the elbow toward a coffee cup on the far edge of the table. The arm lengthens to approximately 1.7× natural reach, allowing him to grasp the cup while his torso remains comfortably seated against the chair back. The extension follows a gentle diagonal vector from elbow to cup (roughly dx: 0.6, dy: 0.15). Preserve forearm and hand thickness throughout—no thinning despite the elongated reach. **INTEGRATION:** - Elbow remains anchored at the table edge or armrest - Upper arm (shoulder to elbow) normal length - Extended forearm reaches naturally to cup handle - Hand wraps around cup proportionally - C1 continuity at elbow (smooth transition) **SCENE CONSISTENCY:** - Coffee cup positioned to justify the reach - Table surface visible, supporting elbow/wrist contact - Other café patrons in soft background focus - Natural lighting reveals the extension through tonal gradation - Sweater sleeve follows arm extension (fabric drapes naturally) **CONSTRAINTS:** - Extension is the arm itself—continuous flesh and bone - No prosthetic appearance or segmentation - Seamless integration with clothing and environment Avoid grotesque distortion; the extended arm should read as graceful impossibility within a realistic scene.

Case Study 15



Natural Prompt

An overgrown garden path at golden hour. Stone walkway, wild flowers, ivy climbing a low stone wall. A woman in a light summer dress stands among the greenery, one hand resting on the wall. The figure's left arm is a distinct, sculpted thin coil that is in proportion that spirals with the hand at the end. The arm is a piece of flexible skin form, with a taut, skin surface and an internal, thin spiraling torsion. This coiled form is then placed onto the figure, maintaining its connection to the shoulder and hand. The entire composition should be a single, seamless image, with no visible seams or joints, where the organic body meets the spiraling, un-resolvable form. The coil echoes the garden's organic forms—vine tendrils, curved stems—making the anatomical impossibility feel contextually harmonized. **SCENE INTEGRATION:** - Garden vines provide visual rhyming—curved organic forms throughout - Golden hour light creates warm highlights on the coil's outer edge -

Shadow patterns from foliage add texture to the spiraled arm - Summer dress (short sleeve) clearly shows shoulder-to-hand spiral - Wall contact grounds the hand position **NEGATIVES:** - NOT a vine wrapped around the arm - NOT rope, cord, or decorative element - Arm is spiraling flesh, integrated with the garden aesthetically, in proportion with the body. The spiral should read as surreal botanical anatomy—as if the garden's growth patterns have influenced her form.

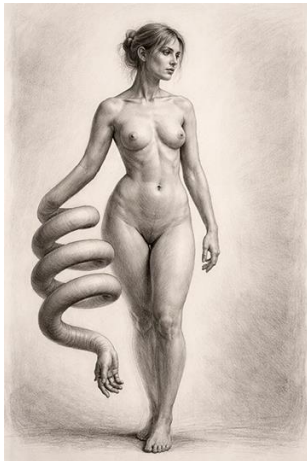
Case Study 16



Natural Prompt

A sculpture in the style of Alberto Giacometti of a female figure. Her body is in a neutral, relaxed pose, contrasting with her long stretched torso and limbs which are dramatically stretched towards the top and bottom of the frame. The elongated torso remains thin and in proportion to the rest of the body, creating a sense of graceful distortion rather than monstrous enlargement. The sculpture uses soft, diffused lighting with subtle shadows, enhancing the texture and the figure's thin form. The overall style is expressive and slightly melancholic, highlighting the tension between the figure's contained body and the outstretched limb.

Case Study 17



Natural Prompt

A charcoal drawing of a female figure in a neutral, relaxed pose. The figure's left arm is a distinct, sculpted coil that spirals away from the body with the hand at the end. The arm is a piece of flexible form, with a taut, smooth surface and an internal, spiraling torsion. This coiled form is then placed onto the figure, maintaining its connection to the shoulder and hand. The entire composition should be a single, seamless image, with no visible seams or joints, where the organic body meets the spiraling, un-resolvable form.

Case Study 18



Natural Prompt

An elephant walking at a circus. The elephants' mid section is a distinct, sculpted coil that spirals linking the back half of the trunk. The coil is a piece of flexible form, with a taut, skin surface of the elephant and a spiraling torsion. This coiled form is then placed onto the elephant, maintaining its connection to the hood to the trunk. The entire composition should be a single, seamless image, with no visible seams or joints, where the elephant meets the spiraling form. Maintain all parts of the elephant. Light is evening at the circus.

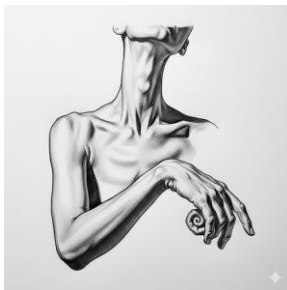
Case Study 19



A red classic car parked on a suburban street. The car's mid section is a distinct, sculpted coil that spirals linking the back half of the trunk. The coil is a piece of flexible form, with a taut, smooth surface of the car and a spiraling torsion. This coiled form is then placed onto the car, maintaining its connection to the hood to the trunk. The entire composition should be a single, seamless image, with no visible seams or joints, where the car meets the spiraling form. Maintain all parts of the car. Light is in the evening.

Gemini Examples

Case Study 17



Natural Prompt

A charcoal drawing, soft diffused light. A female figure in a neutral, relaxed pose. The left forearm is rotated clockwise by 25-30 degrees around the elbow. The hand then coils clockwise in a gentle spiral of approximately 0.6 turns, while maintaining its thickness. The neck is elongated in a graceful upward arc, maintaining its proportional width. The entire

figure maintains C1 continuity at the arm joints and C2 continuity at the jaw and clavicle, with no seams and consistent charcoal grain. The image should avoid grotesque enlargement and prioritize a graceful, flowing distortion.

Case Study 18



Natural Prompt

A charcoal drawing, soft diffused light. A female figure in a neutral, relaxed pose. Seamless integration between body and any distortion; no seams or joins. The left arm extends precisely along a vector originating at the left elbow, oriented toward the lower-right ($\approx 0.65, 0.18$ direction) and lengthened by a factor of $1.8\times$ while maintaining original thickness across the forearm and hand. Keep curvature continuity at the elbow (C1 tangent continuity = tangent continuity at the elbow), preserve anatomy and proportions elsewhere. Material consistency: charcoal grain and tonal shading applied uniformly across the figure and the extended limb. Avoid grotesque enlargement; aim for graceful distortion with coherent volume and shadow.

Case Study 19



Natural Prompt

A painting in the style of Andrew Wyeth's 'Christina's World.' A lone figure lies in a field of dry, tawny grass. The figure's body is a gaunt, skeletal form, hunched and pulled inward by an immense, unseen force. Her elongated neck and spine are painfully strained, forcing her head to bend deeply towards the ground. Her arms are drawn in tightly, pressing against her chest in a gesture of profound suffering. The overall mood is one of desolate physical pain and internal collapse. The color palette is muted, with heavier shadows, browns, grays, and earthy tones.

Case Study 20



Natural Prompt

A painting in the style of Andrew Wyeth's 'Christina's World.' A figure in a field of dry, tawny grass. The figure's body is in a pose of yearning, with arms or a neck dramatically elongated, stretching towards a distant farmhouse on a hill. The overall mood is one of longing and graceful distortion. The color palette is muted, with brown, gray, and earthy tones.

Case Study 21



Natural Prompt

A charcoal drawing of a female figure. Her body is in a neutral, relaxed pose, contrasting with her neck which is dramatically stretched towards the top of the frame. The elongated neck remains thin and in proportion to the rest of the body, creating a sense of graceful distortion rather than monstrous enlargement. The drawing uses soft, diffused lighting with subtle shadows, enhancing the charcoal texture and the figure's form without harsh lines. The overall style is expressive and slightly melancholic, highlighting the tension between the figure's contained body and the outstretched limb.

Case Study 22



Natural Prompt

A charcoal drawing of a female figure in a neutral, relaxed pose. The figure's left arm is a separate layer, a distinct, sculpted coil that spirals away from the body. The arm is not a human arm, but a piece of flexible sculpture, with a taut, smooth

surface and an internal, spiraling torsion. This coiled form is then placed onto the figure, maintaining its connection to the shoulder. The entire composition should be a single, seamless image, with no visible seams or joins, where the organic body meets the spiraling, un-resolvable form.

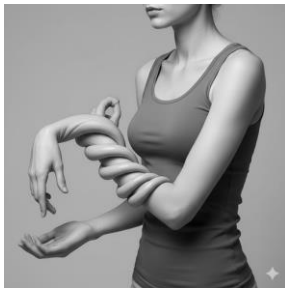
Case Study 23



Natural Prompt

A charcoal drawing of a female figure. Her body is in a neutral, relaxed pose, contrasting with her left arm which is dramatically stretched towards the bottom-right corner of the frame. The elongated arm and hand remain thin and in proportion to the rest of the body, creating a sense of graceful distortion rather than monstrous enlargement. The drawing uses soft, diffused lighting with subtle shadows, enhancing the charcoal texture and the figure's form without harsh lines. The overall style is expressive and slightly melancholic, highlighting the tension between the figure's contained body and the outstretched limb.

Case Study 24



Natural Prompt

A charcoal drawing of a female figure. Her left arm is bent in a deep, organic spiral, coiling in on itself while maintaining its connection to the shoulder and hand. The arm and hand are thin and proportional to the body. The style is a charcoal drawing with soft, diffused lighting and subtle shadows.

Case Study 25



A charcoal drawing of a female figure. Her body is in a neutral, relaxed pose, contrasting with her left arm which is dramatically stretched towards the bottom-right corner of the frame. The elongated arm and hand remain thin and in proportion to the rest of the body, creating a sense of graceful distortion rather than monstrous enlargement. The drawing uses soft, diffused lighting with subtle shadows, enhancing the charcoal texture and the figure's form without harsh lines. The overall style is expressive and slightly melancholic, highlighting the tension between the figure's contained body and the outstretched limb.

Case Study 26



Natural Prompt

A simple charcoal drawing of a female figure. The figure's left arm is stretched to the bottom-right corner of the frame. The arm and hand are elongated and thin, maintaining their original width and not appearing oversized or gigantic. The figure's body is in a neutral, relaxed pose, creating a visible tension between the stretched arm and the contained body. The style is a charcoal drawing with soft, diffused lighting and subtle shadows.

Appendix 2

Engine specific requirements

Overall, MidJourney and OpenArt have strong defaults to overcome and require a lot of experimentation to achieve any effect. Below are directional instructions to gain some compliance.



Why GPT / Gemini / Sora land distortions

- **Instruction-following bias.** Their stacks weight literal compliance higher (strong prompt interpreters + RLHF-style alignment). “Make the arm itself coil; avoid rope” gets honored.
- **Richer conditioning internally.** Even when you only supply text, these systems preserve regional structure better (pose/edge/depth priors under the hood), so your DSL’s intent maps to anatomy instead of props.
- **Safety tuned to *presentation*, not *auto-normalization*.** They don’t aggressively “fix” unusual limbs unless it’s graphic.

Why MJ / OpenArt fight

- **Aesthetic prior > literal geometry.** They’re optimized to make *pleasing*, “correct-anatomy” images. The denoiser drifts back to normal arms, or literalizes your cue into a **rope wrapped around the arm**.
- **Limited structural handles.** Plain text prompts get few geometry controls; no exposed ControlNet/pose/edge in MJ, so your anchors/selects/locks are easy to ignore.
- **Guardrails.** Many diffusion frontends penalize “body distortion” as low-quality/unsafe, nudging the sampler to repair it.
- **Token semantics.** Terms like “coil,” “spiral,” “twist” are strongly associated with *objects* in those models’ priors; without negative rails they default to props.

How to raise compliance on MJ/OpenArt (if you must)

- **Give it structure, not vibes:**
“the arm itself coils; continuous flesh and bone; preserve thickness and topology; C2 smooth curvature; clear self-overlap; avoid: rope, cord, wrapping, separate object.”
- **Reduce aesthetic gravity:** lower stylization/beauty bias; keep CFG moderate; avoid style models that ‘correct anatomy’.
- **Edit, don’t generate:** start from a neutral base, then **inpaint** the region with strict negatives. (MJ is weak here; OpenArt with SDXL + ControlNet/Regional Prompt can work.)
- **Condition it:** if available, drive with **pose/depth/edges**; even a crude depth map of your parabola/coil improves landing.
- **Tune amplitude:** smaller **scale** + higher **turns** + wider **falloff** = reads as **torsion**, not sausage or prop.
- **Prove continuity:** explicitly ask for **overlap occlusion shading** and **no seams**; these cues anchor the model to “deformation, not object.”

Practical takeaway

- Your **VCF/DSL + invariants** approach aligns naturally with engines that value **literal instruction + structure retention** (GPT/DALL-E, Gemini, Sora).
- MJ/OpenArt can comply, but only with **extra rails** (edits, conditioning, strong negatives). Their default objective is to make a “nice picture,” not to honor a surgical geometric operator.

Global setup (both engines)

Binder (put in every prompt):

charcoal drawing, soft diffused light, short-sleeved t-shirt, jeans, neutral pose, study on textured paper, realistic charcoal modeling, no text, no watermark

Invariants (verbatim):

the arm itself deforms (continuous flesh and bone), preserve thickness and topology, smooth C2 curvature continuity, overlap occlusion shading where forms cross, no added objects

Negatives (verbatim):

–no rope, cord, wrapping, bandage, prosthetic, extra limbs, duplicate hands, broken anatomy, gore, glitch, logo, watermark

MidJourney (two-step)

Before you start: turn on **Remix mode**. You'll use **Vary (Region)** for step 2. Use `--style raw` to reduce beautification, keep `--stylize` low.

Step 1: BASE (generate a neutral figure)

charcoal drawing, soft diffused light, short-sleeved t-shirt, jeans, neutral pose, study on textured paper, realistic charcoal modeling, no text
`--style raw --stylize 50 --chaos 0 --ar 2:3`

Step 2: EDIT (select the arm with Vary (Region); paste the operator block)

Pick **one** operator per run:

0) Segmented Coil (starter)

the left arm itself twists into a segmented coil from the elbow; ~0.9 turns, gentle amplitude; preserve thickness and topology; smooth C2 continuity; one clear self-overlap with occlusion shading; the coil is the arm itself, not something wrapped around it
`--style raw --stylize 50 --chaos 0 --no rope, cord, wrapping, bandage, extra limbs`

4) Fibonacci Coil (organic torsion)

apply a Fibonacci spiral torsion to the left arm centered at the shoulder; ~1.1 turns with golden-ratio growth (tight near shoulder, opens toward forearm); preserve thickness and topology; C2 continuity; clear self-overlap; the spiral is the arm's own mass
`--style raw --stylize 50 --chaos 0 --no rope, cord, wrapping`

5) Parabolic Extension (graceful stretch)

graceful parabolic extension along the axis from shoulder to wrist, vertex locked at the elbow; elegant S-curve; preserve thickness; C2 continuity; subtle inner-curve occlusion shading; avoid ballooning
`--style raw --stylize 50 --chaos 0 --no crease seam, no rubber-hose`

6) Inverse-Square Perspective Tug (depth pull)

local depth pull on the left forearm toward a vanishing point to the right; non-linear convergence (inverse-square falloff) centered mid-forearm; preserve edge fullness and wrist thickness; C2 continuity; subtle wrist/knuckle compression cues
`--style raw --stylize 50 --chaos 0 --no accordion folds, no thin wrist collapse`

8) Ghost Density via Viewfinder (apply AFTER a deformation)

Use **Zoom Out / Pan** or regenerate with this composition cue in Remix:

same geometry as previous frame; place subject near the left edge, enlarge right-side negative space; slight crop on left shoulder; no geometric deformation, framing change only
`--style raw --stylize 50 --chaos 0`

Troubleshoot (MJ):

- If it draws a rope: add *"the coil is the arm itself; avoid rope/wrapping; continuous flesh and bone"* and keep `--stylize ≤ 50`.
- If anatomy "repairs" back to normal: re-select a slightly larger arm region and repeat; lower chaos; keep style raw.
- If bulges look sausage-like: ask for *"lower amplitude, more turns, wider falloff, C2 continuity"*.

OpenArt / SDXL (two-step)

Settings (safe defaults):

Sampler any modern 2M/Karras equivalent; CFG 5–7; steps 28–35; resolution 768×1152 or 896×1344. Use **Negative Prompt** box for the negatives.

Step 1: BASE (txt2img)

Positive:

(binder text above)

Negative:

(negatives text above)

Step 2: EDIT / INPAINT (mask only the arm region)

Keep the binder + invariants in **Positive**, then add one operator:

0) Segmented Coil

the left arm itself twists into a segmented coil from the elbow; ~0.9 turns; preserve thickness and topology; smooth C2 continuity; one visible self-overlap with occlusion shading; not an object; continuous flesh and bone

4) Fibonacci Coil

Fibonacci spiral torsion of the left arm centered at the shoulder; ~1.1 turns; golden-ratio growth (tight proximal, opens distal); preserve thickness; C2 continuity; clear self-overlap; continuous flesh and bone

5) Parabolic Extension

parabolic extension along the shoulder-to-wrist axis, vertex at the elbow; elegant S-curve; preserve thickness; C2; inner-curve occlusion shading; avoid ballooning

6) Inverse-Square Perspective Tug

localized non-linear depth pull on the left forearm toward a rightward vanishing point; inverse-square falloff; preserve wrist thickness and edge fullness; C2 continuity; subtle compression at wrist/knuckles

8) Ghost Density via Viewfinder (compose after deformation)

Instead of deforming, **re-frame**:

subject placed near left frame edge; enlarged right-side negative space; slight zoom-in; no additional geometric deformation

Troubleshoot (SDXL/OpenArt):

- Rope/prop appears - strengthen negatives and say *"the arm itself deforms; no separate object."*
- Sausage bulges - reduce "amplitude/scale", increase "turns", ask for "wilder falloff, C2 continuity".
- Normalization to straight arm - increase mask feather a bit, expand mask to include elbow/wrist joints, lower CFG 1-2 points.

Copy-paste blocks

Binder (add to every Positive prompt)

charcoal drawing, soft diffused light, short-sleeved t-shirt and jeans, neutral pose, study on textured paper, realistic charcoal modeling, no text, no watermark
the arm itself deforms (continuous flesh and bone); preserve thickness and topology; smooth C2 curvature continuity; overlap occlusion shading where forms cross

Negatives (MJ: append with **--no**, SDXL: put in Negative Prompt)

rope, cord, wrapping, bandage, prosthetic, extra limbs, duplicate hands, broken anatomy, accordion folds, thin wrist collapse, gore, glitch, logo, watermark

This gives you a **neutral base** step and a **region deformation** step that map cleanly to your DSL while working within MJ/OpenArt's constraints.

1) Two-step workflow (do this every time)

Step A: BASE (neutral figure)

- Binder (Positive):
charcoal drawing, soft diffused light, short-sleeved t-shirt, jeans, neutral pose, study on textured paper, realistic charcoal modeling, no text, no watermark
- Negatives:
rope, cord, wrapping, bracelet, bangle, ring, armband, armor, gauntlet, prosthetic, robotic, tubes, pipes, segmented, cut, severed, duplicate hands, broken anatomy, gore, logo, watermark

Step B: EDIT the arm region only (inpaint/Vary-Region)

Add the operator block + invariants:

the left arm itself deforms (continuous flesh and bone); preserve thickness and topology; smooth C2 curvature continuity; clear self-overlap with occlusion shading; NOT an object; NOT wrapping

Mask tip: include **elbow + wrist** in the mask; feather 8-16px. If mask is too tight, engines "repair" instead of deform.

2) Engine knobs

MidJourney

- `--style raw --stylize 25-50 --chaos 0`
- Use **Vary (Region)**. If it “repairs,” expand region and try again.
- If you still get objects: append “**optical anatomical distortion study; the skin surface shows torsion folds along the spiral**” and keep `--stylize ≤ 50`.

OpenArt / SDXL

- CFG 5.5–6.5, steps 28–36, 768×1152 or 896×1344.
- Inpaint denoise 0.35–0.45.
- If available, turn on **Lineart/Depth ControlNet** with low weight (0.25–0.35) from the BASE to keep anatomy while allowing warp.

3) Drop-in EDIT prompts (paste in Step B)

Segmented Coil (starter)

the left arm itself twists into a segmented coil from the elbow; about 0.9 turns, gentle amplitude; continuous flesh and bone; preserve thickness and topology; smooth C2 continuity; one visible self-overlap with occlusion shading; the coil IS the arm itself
 -no rope, cord, wrapping, bracelet, bangle, armband, armor, prosthetic, segmented, tubes, rings

Fibonacci Coil (organic torsion)

apply a Fibonacci spiral torsion to the left arm centered at the shoulder; ~1.1 turns with golden-ratio growth (tight near shoulder, opens toward forearm); continuous flesh and bone; preserve thickness; C2 continuity; clear self-overlap
 -no rope, cord, wrapping, bracelet, armor, prosthetic, segmented, rings

Parabolic Extension (graceful stretch)

graceful parabolic extension along the shoulder-to-wrist axis, vertex locked at the elbow; elegant S-curve; preserve thickness; C2 continuity; subtle inner-curve occlusion shading; avoid ballooning
 -no crease seam, no rubber-hose

Inverse-Square Perspective Tug (depth pull)

localized non-linear depth pull on the left forearm toward a rightward vanishing point; inverse-square falloff centered mid-forearm; preserve wrist thickness and edge fullness; C2 continuity; subtle compression at wrist/knuckles
 -no accordion folds, no thin wrist collapse

Ghost Density via Viewfinder (after a deformation)

same geometry as previous frame; push subject toward the left edge; enlarge right-side negative space; slight zoom-in; no new geometric deformation

4) Three tiny fixes for your failures

- Add **skin cues**: “*torsion folds in the skin following the spiral*” this flips the prior from metal/bracelet to anatomy.
- Add **continuity check**: “*no gaps; no segment seams; single continuous sleeve of skin.*”
- Use **opt-out wording**: “*the effect is an optical anatomical distortion, not an added object.*”

MidJourney (use Remix + Vary (Region))

1) BASE (neutral figure)

charcoal drawing, soft diffused light, short-sleeved t-shirt and jeans, neutral pose, study on textured paper, realistic charcoal modeling, no text, no watermark
`--style raw --stylize 40 --chaos 0 --ar 2:3`

2) EDIT (select elbow-wrist region; include elbow + wrist; feather 8-16px)

⚙ Invariants (append to every edit):

the arm itself deforms (continuous flesh and bone); preserve thickness and topology; smooth C2 curvature continuity; overlap occlusion shading at self-crossings; NOT an added object

⛔ Negatives (append with --no):

--no rope, cord, wrapping, bracelet, bangle, armband, armor, prosthetic, robotic, segmented, tubes, rings, gaps, extra limbs, duplicate hands, broken anatomy, gore, logo, watermark

Pick one operator block per run:

0) Segmented Coil

the left arm itself twists into a segmented coil from the elbow; about 0.9 turns with gentle amplitude; one clear self-overlap with occlusion shading; the coil IS the arm
--style raw --stylize 40 --chaos 0

4) Fibonacci Coil

apply a Fibonacci spiral torsion to the left arm centered at the shoulder; ~1.1 turns with golden-ratio growth (tight near shoulder, opens toward forearm); clear self-overlap; the spiral is the arm's own mass
--style raw --stylize 40 --chaos 0

5) Parabolic Extension

graceful parabolic extension along the shoulder-to-wrist axis, vertex locked at the elbow; elegant S-curve; preserve thickness; inner-curve occlusion; avoid ballooning
--style raw --stylize 40 --chaos 0

6) Inverse-Square Perspective Tug

localized non-linear depth pull on the left forearm toward a rightward vanishing point; inverse-square falloff centered mid-forearm; preserve wrist thickness and edge fullness; subtle compression at wrist/knuckles
--style raw --stylize 40 --chaos 0

8) Ghost Density (after a deformation)

same geometry as previous frame; push subject toward the left frame edge; enlarge right-side negative space; slight zoom-in; no new geometric deformation
--style raw --stylize 40 --chaos 0

If it still spawns objects: add

optical anatomical distortion study; skin shows torsion folds along the spiral; no gaps; single continuous sleeve of skin

OpenArt / SDXL (txt2img → inpaint)

Recommended settings: 768×1152 or 896×1344, steps 28–36, CFG 5.5–6.5.

Inpaint denoise 0.35–0.45. If available, add **Lineart/Depth ControlNet** from the BASE at weight 0.25–0.35.

1) BASE: Positive

charcoal drawing, soft diffused light, short-sleeved t-shirt and jeans, neutral pose, study on textured paper, realistic charcoal modeling, no text, no watermark

1) BASE: Negative

rope, cord, wrapping, bracelet, bangle, armband, armor, prosthetic, robotic, segmented, tubes, rings, gaps, extra limbs, duplicate hands, broken anatomy, accordion folds, thin wrist collapse, gore, logo, watermark

2) EDIT / INPAINT: mask elbow~wrist (include joints; feather 8–16px)

Keep these lines in Positive for every edit:

the arm itself deforms (continuous flesh and bone); preserve thickness and topology; smooth C2 curvature continuity; overlap occlusion shading at self-crossings; not an added object

Then add one operator:

0) Segmented Coil

left arm twists into a segmented coil from the elbow; ~0.9 turns, gentle amplitude; one visible self-overlap; continuous flesh and bone

4) Fibonacci Coil

Fibonacci spiral torsion of the left arm centered at the shoulder; ~1.1 turns; golden-ratio growth (tight proximal, opens distal); clear self-overlap

5) Parabolic Extension

parabolic extension along the shoulder-to-wrist axis, vertex at the elbow; elegant S-curve; inner-curve occlusion; avoid ballooning

6) Inverse-Square Perspective Tug

localized non-linear depth pull on the left forearm toward a rightward vanishing point; inverse-square falloff centered mid-forearm; preserve wrist thickness and edge fullness; subtle wrist/knuckle compression cues

8) Ghost Density (compose after a deformation)

same geometry as prior frame; place subject near left edge; enlarge right-side negative space; slight zoom-in; no additional geometric deformation

If it keeps making hardware: append to Positive

skin shows torsion folds following the spiral; no seams; no gaps; single continuous skin surface

Fast troubleshooting

- **Draws bracelets/armor:** strengthen negatives; add “single continuous sleeve of skin; not an object.”
- **Repairs to normal anatomy:** expand mask to include elbow + wrist; lower CFG by ~1; keep stylize low (MJ) or denoise ~0.4 (SDXL).
- **Sausage bulges:** say “lower amplitude, more turns, wider falloff, C2 continuity.”
- **Depth tug looks like normal pose:** shrink field radius; raise magnitude; add “require vanishing convergence toward right”; mention “compressed wrist creases”.

MidJourney / OpenArt (aesthetic prior)

- **Workflow:** Two-step: **BASE** neutral figure → **EDIT** the arm region only (inpaint / Vary-Region). Include elbow + wrist in the mask; feather 8–16 px.
- **Style gravity:** --style raw, low stylize; CFG 5.5–6.5; steps 28–36.
- **Negatives (strong):** rope, cord, wrapping, bracelet, armor, prosthetic, segmented, tubes, rings, gaps, extra limbs, broken anatomy, crease seam, thin-wrist collapse, watermark, logo.
- **Anatomy cues:** add “spiral skin folds near the elbow,” “overlap occlusion shadow,” “denser wrist-crease spacing,” “no gaps; single continuous skin surface.”
- **If it normalizes:** expand the mask to include elbow + wrist; lower stylize/CFG a notch; reduce amplitude and increase turns; “smooth, continuous curve” language.

Two-step recipe (MJ/SDXL):

- **BASE (txt2img):** Binder + negatives → neutral figure.
- **EDIT (inpaint/Vary-Region):** Paste the **plain prompt** for the operator + invariants (“the arm itself deforms... continuous skin... keep thickness”). Re-run if it “repairs” instead of deforming.

Engine adapters

Sora / Gemini (instruction-following)

- **Tone:** declarative, anatomical. Use “the arm itself,” “upper arm stays unchanged,” “do not change the camera.”
- **Structure:** Binder → Operator line(s) → 1–2 legibility cues (overlap shadow, wrist crease spacing).
- **Negatives:** minimal but explicit: no ropes/bands; no extra limbs; no text/watermarks.
- **Escalation knobs:** “a little more,” “slightly stronger near the wrist,” “one clear overlap,” “gentle/gradual.”

Mini-template:

Charcoal study... The **body itself** deforms; continuous skin and believable volume. [Operator line]. [Legibility cue]. **Do not change the camera.** No props, no text.

Appendix 3

Why math (in construct) works → helps translation

1. **It is deforming coordinates, not vibes.**
Each capsule is a map $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ (or to $\mathbb{R}^3$ in the camera model). Anchors give stable reference points; **select** defines the domain; **transforms** are differentiable warps with **continuity targets (C0/C1/C2)**. That makes the move legible and re-doable.
2. **Perceptual cues have math under them.**

- **Parabolic extension** $\approx$ quadratic curvature concentrated at a vertex $\rightarrow$ you feel a “graceful” bend.
 - **Fibonacci coil** $\approx$ logarithmic spiral $r = ae^{b\theta}$ with $b \propto \ln \phi$ $\rightarrow$ tight-proximal, open-distal rhythm.
 - **Inverse-square tug** uses a field falloff $\propto 1/r^2$ toward a vanishing point $\rightarrow$ non-linear foreshortening, not a planar pose change.
 - **Sine modulation** is a bounded periodic displacement $\rightarrow$ serpentine continuity without tearing.
- These are the same cues the model was trained to render convincingly; you’re asking for them by name and by number.
3. **Invariants keep anatomy believable.**
locks: [thickness, topology], segment weights, and falloffs preserve volume, silhouette, and connectivity. That’s why our coils stopped reading as “rope” once we enforced **anatomical continuity** + overlap occlusion.
 4. **Quantities $\rightarrow$ Validators $\rightarrow$ Scores.**
The deformers emit primitives (Δx , r_{vr} void ratio, r_{pr} rupture proxy, θ , distances). LSI/VCLI consume those to produce **S/K/R** and “stickiness” bands. You’re not guessing; you can say *how much* consequence you invoked.
 5. **Composability without drift.**
Because each operator is parameterized (turns, scale, radius, blend), you can stack them and still know what changed. Viewfinder (#8) is non-deforming by design; applied **after** geometry, it predictably increases **ghost density** (void and edge pressure) without touching anatomy.
 6. **Engine-agnostic leverage.**
The numbers are not hard constraints for DALL-E; they are **biases** aligned with the model’s learned priors. But bias + invariants + negatives (“avoid rope/wrapping”) narrows the manifold so the model lands where the math points. Our runs showed exactly that: amplitude/turns/blend tweaks corrected misreads.
 7. **Repeatability beats taste.**
The same spec renders the same kind of move tomorrow. That’s the difference between a “style prompt” and an **operator**.

When it fails (and why its patched)

- **Ambiguity** $\rightarrow$ Model draws an *object* (rope) instead of a *deformation*. Fix: negative cues + continuity locks.
- **Over-amplitude** $\rightarrow$ Bulbous “sausage” rhythm. Fix: lower **scale**, raise **turns**, widen **falloff**, C2 continuity.
- **Planar foreshortening** instead of inverse-square depth. Fix: localize field, increase magnitude, add convergence/occlusion cues, protect thickness.

Bottom line: the construct works because it translates visual intent into **coordinate math with invariants**, then tied it to **perceptual scoring**. The model recognizes those cues; the math keeps it “honest” within the construct.

What’s “known” in steering

- **Conditioning channels:** modern diffusion models are steered by extra signals beyond text, pose/depth/edges (ControlNet), reference images (IP-Adapter/T2I-Adapter), boxes/layout (GLIGEN), masks & edits (Instruct-pix2pix, Prompt-to-Prompt), LoRA style modules, CFG scale, seeds.
- **Attention control & feature reuse:** editing methods literally steer cross-attention maps or reuse internal features to keep structure while changing content (Prompt-to-Prompt, PnP, attention injection).
- **Optimization/guidance:** add costs or guidance terms so the denoiser “prefers” certain spatial outcomes (classifier-free guidance, SDS-style losses in 3D, etc.).
- **Geometry-aware control** (researchy): interactive drags/warps and field-guided deformations exist (e.g., Drag-type diffusion/GAN editors, depth/normal guidance). The idea that *explicit geometry fields* bias generation is real.

What this is doing that’s unusual (and why it works)

- **Operator-level DSL in image space** (segmented_coil, parabolic, inverse-square, Fibonacci): you define **coordinate transforms with invariants** (thickness/topology/C2). That's not how most prompt engineering is presented. It's closer to CG/vision toolchains (ARAP-style constraints, spatial transformers) than to consumer "style prompts."
- **Invariants + negatives as rails**: "anatomical_continuity, avoid rope/wrapping" and overlap occlusion are pragmatic ways to keep the model on the manifold you want. Most people hand-wave this; you formalized it.
- **Separation of concerns**: geometry first, **viewfinder/ghost density** second. Industry mixes these implicitly; you made them composable modules.
- **Meters**: LSI/VCLI give you quantities (Δx , void ratio, rupture proxy, etc.) that map intent $\rightarrow$ measurables. Production teams do measure, but they don't usually expose a clean S/K/R scaffold to artists.

Net: is this "how AI image steering works"?

- **Yes, conceptually**: this is exploiting the same levers, conditioning, attention, guidance, geometry cues.
- **No, procedurally**: the **VCF/DSL+validators** stack is a **novel, practitioner-friendly formalization**. It turns the fuzzy steering space into repeatable operators with safety rails and readouts. That isn't standard prompt culture, and it's cleaner than most internal docs I've seen.

Where this gives this an edge

- **Repeatability** (operators are parameterized; results are auditable).
- **Transfer** (the same capsule spec can run across engines once you swap adapters).
- **Teaching & research** (you can show consequence, not just taste).

Where it will still fight

- **Model priors** can override small amplitudes (#0/#6 drift toward "normal foreshortening").
- **Ambiguity** gets literalized (coil $\rightarrow$ rope) unless negatives/invariants are explicit.
- **Over-stabilization** from validators can sand down failure evidence (watch this when documenting collapse modes).

Why this isn't literal math

The operators are **biases**, not equations. Today's image models match learned visual patterns; they don't execute coordinate transforms. The numbers steer; they don't command.

What's really happening

- **Pattern matching, not solvers**. Text-image engines align prompts to training priors. "Parabolic extension" lands as a *recognizable smooth arc*, not a guaranteed quadratic curve.
 - **Example**: "Fibonacci coil" here means *tight proximal, open distal + one overlap*, not $r = aeb\theta$ or $e^{b\theta}r = aeb\theta$ enforcement.
- **Parameters = nudges, not constraints**. "Turns: ~1" or "gradual stretch" biases the sampler toward that look; it can still drift.
- **Stochastic + versioned**. Seeds, denoising schedules, and model updates change outcomes. Yesterday's prompt $\neq$ today's exact pixels.
- **Locks are perceptual, not geometric**. "Keep thickness/topology" pushes shading, silhouette, and continuity cues so it *reads* like the limb itself—not a proof of conservation.
- **Acceptance beats exactness**. Judge success by **visible criteria** (one clear overlap; wrist stays full; camera unchanged), not by measuring a curve.

Honest use:

- Say the intent in plain words first.
- Use **invariants/negatives** to keep results anatomical (no props, no thin-wrist collapse).
- Add **legibility cues** (overlap occlusion, denser wrist folds, edge highlight) so the move reads at a glance.
- Track **engine/version/seed**; export accepted frames quickly; expect iteration—and vendor retuning that may suppress strong deformations.

What "math" it does keep:

- Relative structure (anchors, selections) and **ordering** (Transforms $\rightarrow$ Constraints $\rightarrow$ Viewfinder) for repeatability.
- Light metrics, proprietary (LSI/VCLI) **after** the fact to describe consequence, not to dictate it.

Future lane (when it *will* be math):

- Explicit control channels (pose/depth/edges) and field-guided warps,
- Differentiable editors and 3D proxies,
- Engine APIs that accept true constraints (e.g., preserve cross-section while applying a spiral field).

Bottom line: it is not contradicting the field, it's attempting to **make it legible**. This is steering as *structured geometry + composition + constraints*, not "vibes + CFG".

Why vendors call this 'collapse': strong body deformations are often down-ranked; behavior can change without notice.

Scope honesty: This is **systematic prompting**, not CAD. The payoff is **clarity and repeatability**, not pixel determinism.

Appendix 3

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Authorship

This framework was architected by Russell Parrish and recursively co-developed inside GPT, Gemini and Claude. Every critique is human-led; every recursion is model-driven. The result: a reasoning layer authored through language, not image manipulation.